MIL-P-17840C(SH)
10 March 1986
SUPERSEDING
MIL-P-17840B(SHIPS)
28 December 1962
(See 6.7)

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, CLOSE-COUPLED, NAVY STANDARD (FOR SURFACE SHIP APPLICATION)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven. Navy standard, close-coupled , single-stage, centrifugal pumps.

1.1.1 The Navy standard pump numbers under this specification have the letter "C" added (for example, 9C) to distinguish the pumps covered herein from those under the previous editions of this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
HH-P-46 - Packing; Asbestos, Sheet, Compressed.
QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL (Continued)

QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-288	-	Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
TT-P-645	-	Primer, Paint, Zinc Chromate, Alkyd Type.
GGG-E-950	-	Extractor, Stuffing Box and Pump Packing (and Tamper); and Extractor, Lantern Gland-
GGG-P-781	-	Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-1222	-	Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-P-15024	-	Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	-	Plates, Identification.
MIL-M-15071		Manuals, Technical, Equipment and Systems Content Requirements for.
MIL-E-15090		Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15137		Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
DOD-P-15328		Primer (Wash), Pretreatment (Formula No. 117 for Metals). (Metric)
MIL-C-15345		Castings, Nonferrous, Centrifugal-
MIL-C-15726		Copper-Nickel Alloy, Rod, Flat products (Flat Wire , Strip, Sheet, Bar, and Plate) and Forgings.
MIL-T-16420		Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-P-16789		Packaging of Pumps, Including Prime Movers and Associated Repair Parts.
MIL-M-17059		Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use).
MIL-M-17060		Motors , 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-M-17413		Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-M-17556		Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-C-20159		Copper-Nickel Alloy Castings (UNS No. C96200 and C96400).
MIL-P-24377		Packing Material, Asbestos, Braided, Impregnated with PTFE (Polytetrafluoroethylene), Surface Lubricated.
DOD-G-24508		Grease, High Performance, Multi-Purpose. (Metric)
MIL-S-46163		Sealing, Lubricating, and Wicking Compounds: Thread-Locking, Anaerobic, Single-Component.

STANDARDS

FEDERAL

FED-STD-H28		Screw-Thread Standards for Federal Services.
FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

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MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type 11 - Internally Excited).
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-1399, Section 301 - Interface Standard for Shipboard Systems Ship Motion and Attitude.
DOD-STD-2147 - Technical Repair Standards (TRS); Hull, Mechanical and Electrical (HM&E) Preparation of. (Metric)
MS16142 Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.
MS18229 Plug for "O" Ring Gasket.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

B-153 - Packings and Gaskets, Standard Application of.
B-214 - Root Connections for Attaching Piping.
810-1385850 - Piping, Installation, Pressure for All Services.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 494 - Standard Specification for Castings, Nickel and Nickel Alloy l
A 744 - Standard Specification for Castings, Iron-Chromium-Nickel, Nickel-Base, Corrosion Resistant, for Severe Service. (DoD adopted)
B 61- Standard Specification for Steam or Valve Bronze Castings. (DoD adopted)
B 164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire. (DoD adopted)
B 369 - Standard Specification for Copper-Nickel Alloy Castings. (DoD adopted)
B 446 - Standard Specification for Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) Rod and Bar.
B 584 - Standard Specification for Copper Alloy Sand Castings for General Applications. (DoD adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE
Test Code of Standards - Centrifugal Pump Section.

(Application for copies should be addressed to Hydraulic Institute, 712 Lakewood Center N., 14600 Detroit Avenue, Cleveland, OH 44107.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications , specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General design. The design and construction of pumps shall comply with the space and weight requirements as specified (see 6.2.1). The pump unit shall perform under the list, pitch, roll, and trim conditions as specified in MIL-STD-1399, section 301. Pumps shall operate continuously at any point over the entire flow range and the design conditions as specified (see 6.2.1).

3.2.1 Reliability. The requirements of this specification are imposed exclusively to obtain equipment of utmost reliability for the service intended. The assurance of maximum reliability shall be the paramount controlling principle in the design, fabrication, assembly, and testing of this equipment. Users of and contractors to this specification are urged to communicate to NAVSEA any findings related to the requirements or lack of requirements of this specification whereby improvement of equipment reliability can be achieved.

3.2.2 The equipment specified herein will be operated, maintained, and repaired on board Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, to the fact that maintenance and repair will be made underway in heavy seas, that equipment will operate unattended, and that maintenance personnel may not be seasoned mechanics. The requirements for maximum reliability directly relate to those shipboard environmental and service conditions, and they shall be fully considered in the pump design. The aspects of “human engineering” shall also be fully explored and considered in the equipment design to minimize the possibility of failure through improper operation and maintenance and to preclude personnel safety hazards. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable functionally and from considerations of strength.

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3.2.2.1 The equipment shall have maximum repair accessibility for ease of examination of wearing parts and for simplicity of disassembly and proper reassembly. Positioning and alignment of parts in assembly shall employ positive means such as shoulder, tongue and groove, or other locating techniques whereby correct reassembly is repeatedly assured. In consonance with the concept of ease of maintenance and where not already specified, the pump designer shall consider all of the following features for incorporation in the design and technical documentation wherever such incorporation will significantly contribute to ease of maintenance without compromising performance reliability:

(a) Lifting lugs.
(b) Casing assembly guide pins.
(c) Alignment and positioning dowels.
(d) Jacking screws.
(e) Ease of access to coupling.
(f) Re-makeable piping connections.
(g) Piping out of way of maintenance access.
(h) Ease of packing, mechanical seal, bearing, and wearing ring replacement.
(i) Split casing, bearing housings, and so forth.
(j) Guards for rotating and moving components.
(k) Warning plates to prevent casualties to equipment and personnel.
(l) Provisions for connecting instruments for performance evaluation.
(m) Attached instruments for monitoring performance.
(n) Visibility and access to attached instruments.
(o) Thoroughness of operating instructions.
(p) Thoroughness of preventive maintenance instructions.
(q) Explicit assembly and disassembly instructions.
(r) Exploded views of critical assemblies.
(s) Adequate system diagrams.

3.2.2.2 Each pump shall have a service life of 30 years, of which 40 percent shall be actual operation. There shall be no limit on the number of starts during the life of the pump.It shall be assumed that during the life of the pump, parts subject to unavoidable wear and deterioration (with the exception of packing and seals) will be replaced at intervals no shorter than 3 years. The parts subject to wear, deterioration, and normally requiring replacement at 3-year intervals during the service life of the pump (with the exception of packing and seals) shall have a life of 10,000 actual operating hours , and they shall be identified in appropriate drawings and manuals. The requirement set forth above shall not be construed or interpreted as a warranty requirement, nor otherwise affect the manufacturer’s warranty.

3.2.2.3 Maintainability. The time to perform pump maintenance action requiring part removal shall not exceed 3 hours.

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3.2.2.4 In-order to ensure comparability of parts, calculations of the worst-case cumulative effect of dimensions and tolerances for parts and location shall be provided. These calculations shall demonstrate:

(a) There will be no interferences between rotating and stationary parts as-assembled.
(b) There will be no interference between rotating and stationary parts, and hence no reduction in life of wearing parts, when these worst-case cumulative effects are combined with operational, environmental, and thermal loading.

3.2.3 Shock and vibration.

3.2.3.1 Pumps and drivers, all appurtenances and controls shall pass a shock test in accordance with MIL-S-901, grade A, as specified in 4.6.6.

3.2.3.2 Shock mounts (resilient mounts for shock attenuation) shall not be used. All pumps shall pass the specified shock test solidly mounted, whether or not noise attenuation mounts are specified for mounting the- unit in the ship.

3.2.3.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be not more than 1/32 inch larger than the bolts for sizes up to and including 3/4 inch, and no more than 1/16 inch larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 or better of MIL-S-1222.

3.2.3.4 Pumps shall not be damaged or caused to malfunction by either the environmental vibrations or by internally excited vibrations. Units furnished under this specification shall be tested to meet the requirements of MIL-STD-167-1, type I and II (see 6.2.1).

3.2.4 Alignment.

3.2.4.1 The design of the pump units shall be such as to permit handling, shipment, and Installation of the units on board ship without disturbing alignment and such that the normal distortion, weaving, or vibration of the supporting structures on board ships cannot cause fracture or appreciable distortion.

3.2.4.2 Alignment between separate components of a unit shall be maintained by means of keyways, rabbeted or tongue and groove joints, fitted bolts, or other means. Parts manufactured to the same drawings shall be interchangeable.

3.2.4.3 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.4.4 The alignment shall be rechecked by the installing activity after final installation. This can be done by rotating the shaft by hand at least three revolutions.

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3.2.5 Noise. The design, construction, and workmanship of the equipment shall be such that noise levels during operation at the-design conditions will meet the airborne noise and structureborne vibration noise requirements as specified (see 6.2.1 and 4.6.9).

3.2.6 Mounting.

3.2.6.1 Installation. Pumps will be normally installed to operate in a horizontal position; however, pumps shall be capable of operation and mounting in a vertical position with the motor above the pump. Pumps shall be suitable for installation either fore and aft or athwartship (see 6.2.1).

3.2.6.2 Dimension. Pumps shall have mounting dimensions and location of piping connections as shown on figure 1.

3.2.6.3 Mounting brackets. Mounting brackets shall contain a drip packet for the collection of leakage from the stuffing box. The size of the tapped holes shall be as shown on figure 1. Tapped holes shall be tapered thread. Mounting brackets shall be separate from the casings and shall" not include the pump stuffing box or any portion of the pressure boundary.

3.2.6.4 Motors shall be assembled to pumps at the pump manufacturer's plant.

3.2.7 Bearings.

3.2.7.1 Size of radial and thrust ball bearings. The *size* of the radial and thrust ball bearings for each size motor frame shall be as shown on figures 2 and 3. The pump end bearing shall be fixed in the housing and shall carry the pump thrust load. However, on low noise level units, the motor end bearing may carry the thrust load. Bearings shall be selected for a B-10 life of 10,000 hours calculated in accordance with FF-B-171. Calculations shall be made for the design point and also for the highest loading conditions over the operating range and shall consider both thrust and radial hydraulic loads, gravity loads, and bearing preloads. Hydraulic thrust due to the use of single inlet impellers without balancing rings shall be computed in accordance with the following formula:

$$T = P(A_1 - A_0) + 0.05\ P(A_2 - A_1)$$

Where:

T = Maximum unbalanced thrust in pounds.
P = Pressure in pounds per square inch (lb/in^2) at shutoff.
A_2 = Area of impeller at radius R_2 in square inches (see figure 4).
A_1 = Area of impeller to outside of impeller wearing ring, radius R_1 in square inches.
A_0 = Area in square inches to outside of shaft sleeve, radius R_0.

3.2.7.2 When the axial thrust load calculated by the formula of 3.2.7.1 exceeds the permissible thrust loads shown in table I, special thrust balancing devices shall be added.

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TABLE I. Permission thrust loads.

Bearing size	Revolutions per minute (r/rein)			
	1150	1750	2800	3500
305	----	215	----	160
306	----	280	----	215
307	----	235	----	255
308	----	400	----	300
309	560	470	375	350
310	600	500	400	360
312	700	580	470	425
313	800	670	530	495

3.2.7.3 Anti-friction bearings shall be type 111, class 2, or type 120 in accordance with FF-B-171. The sizes used for the various frames shall be as shown on figures 3 and 5.

3.2.7.4 For installations utilizing the pump end bearing as the thrust bearing, a bearing preload spring shall be provided and shall be installed against the outer edge of the outer ring of the front end bearing. For installations using the front end bearing as the thrust bearing, a bearing preload spring shall be installed against the inner edge of the outer ring of the pump end bearing.

3.2.8 Lubrication.

3.2.8.1 The lubrication of rolling contact bearings shall be in accordance with FF-B-171.

3.2.8.2 The rolling contact bearing lubrication method shall prevent bearing failure resulting from over-greasing. Bearing relubrication procedures shall be accomplished with the unit secured.

3.2.8.3 Grease lubricants shall be in accordance with DOD-G-24508. The design shall be such that any qualified grease under DOD-G-24508 can be used interchangeably with any other qualified product to the same specification. Bearings shall be capable of being relubricated without removing the bearings and shall not require frequent lubrication if the pump is not operated over an extended period of time.

3.2.8.4 Thrust bearing housings shall be so designed that the thrust bearing will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by properly locating lubricant supply and drain fittings and by adequate sealing of the bearing housings so that, regardless of length of time the units are idle, an adequate supply of lubricant will remain in the bearing housings.

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3.2.9 Packing and gasket. Packing and gaskets shall be in accordance with Drawing B-153. Packing shall be of the teflon impregnated type in accordance with MIL-P-24377. Gasket compression and tolerances shall not effect bearing bushing or wearing ring fits to an extent that will adversely affect reliability and performance. Pump casing joints shall be made up using compressed asbestos sheet gaskets or 0-rings when approved by the design review agency.

3.2.10 Threaded fasterners. Threaded parts, such as bolts, studs, and nuts, shall conform to FED-ST-H28, FED-STD-H28/2, and MIL-S-1222. The use of tap-bolts or cap-screws is prohibited unless explicitly discussed and justified in a submittal to the design review agency and approved by the design review agency and where the use of through bolts or studs is impractical. For temperatures under 250 degrees Fahrenheit (“F), the setting end of studs shall be a class 2 fit or preferably a class 3 fit, both with locking resin in accordance with MIL-S-46163, type I, grade K. The nut end shall be a class 3 fit.

3.2.10.1 The length of minimum stud engagement shall be computed as specified in FED-STD-H28.

3.2.10.2 Unless otherwise specified (see 6.2.1), threaded fasteners of normally stocked lengths shall be used. Male threads on threaded fasteners, after being installed and tightened, shall protrude at least one thread beyond the top of the nut or plastic lccking element. Excessive protrusion shall be avoided , particularly where necessary clearances, accessibility, and safety are important. Where practicable, the number of threads protruding shall not exceed five; however, in no case shall thread protrusion exceed 10 threads. Washers shall not be used for the sole purpose of lessening thread protrusion.

3.2.11 Piping root connections. Root connections for attaching pipes shall be in accordance with Drawing B-214.

3.2.11.1 Pumps shall be fitted with flanged connections conforming to MIL-STD-777 for suction and discharge pressure gauges or MS connections in accordance with MS16142, as specified (see 6.2.1). All pump flanges shall have a face circular lay finish as detailed in the applicable general note of MIL-STD-777.

3.2.11.2 Pressure gauges, when provided with the units, shall be mounted in accordance with Drawing 810-1385850. Gauges shall be positioned for accessibility of reading and maintenance. Gauges shall be secured on the parent equipment in such a manner as to prevent vibration, breakage, and disconnection.

3.2.12 Welding. Welding and welding repair shall be in accordance with MIL-STD-278. Brazed joints shall not be used.

3.2.13 Mercury prohibition. When specified (see 6.2.1), pumps, drivers, and auxiliary parts furnished by the pump contractor shall be free of mercury contamination and free of functional mercury. The manufacturer shall certify that the pumps, drivers, and auxiliary parts, when shipped, are free from mercury contamination (see 6.2.2).

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3.2.14 Motors.

3.2.14.1 The standardized frame numbers for the various ratings of close-coupled pump motors shall be as specified in tables II and III. Frame flange and shaft extension dimensions shall be as shown on figures 2, 3, and 5, as applicable. The mounting bracket between the motor and pump volute shall be furnished by the pump contractor.

3.2.14.2 Alternating current (ac) motors. Ac motors shall be in accordance with MIL-M-17059 for fractional horsepower (hp) or MIL-M-17060 for integral hp, except as otherwise specified herein, and shall conform to the following classification requirements:

Service	A
Ambient temperature	50 degrees Celsius (°C)
Duty	Continuous
Type	Squirrel-cage induction
Design (for integral hp motors)	B
Bearings	Ball (see figures 2 and 3)
Insulation	Class B, F, H, or N
Temperature rise	70°C in 50°C ambient
Number of phases	3
Frequency	60 Hertz (Hz)
Voltage	440 volts
Enclosure	Totally enclosed or totally enclosed fan-cooled (open motors not permitted)

3.2.14.3 Direct current (dc) motors.

3.2.14.3.1 Dc motors shall be in accordance with MIL-M-17413 for integral hp, or MIL-M-17556 for fractional hp, except as otherwise specified herein, and shall conform to the following classification requirements:

Service	A
Ambient temperature	50°C
Duty	Continuous
Voltage	115 or 230 volts as specified (see 6.2.1)
Winding	Shunt or shunt stabilized
Bearings	Ball (see figures 2 and 5)
Insulation	Class B and H (no silicone materials)
Temperature rise	70°C in 50°C ambient
Enclosure	Totally enclosed or totally enclosed fan-cooled (open motors not permitted)

3.2.14.3.2 Speed tolerance. The full load speed of dc constant speed motors with full FIELD, at ambient temperature (40°C) shall be not less than 85 percent of the specified speed for integral and fractional hp motors. Integral

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hp motors at full load and at operating temperature shall be not less than 92-1/2 percent nor more than 107-1/2 percent of the specified speed: Fractional hp motors at full load and at operating temperature shall be not less than 90 percent nor more than 110 percent of the specified speed.

TABLE II. Ac motors 50°C ambient, class B, F, H, or N insulation with 70°temperature rise (see 3.2.14.2).

Pump	Frame number					
	Hp	3500 r/min	1750 r/min	1150 r/min	AJ 1/	AK 1/
1C-1750-0.33	1/3		56			
1C-3500-0.50	1/2	56	56			
1C-3500-0.75	3/4	56				
1C-3500-1	1	182TCN			7-1/4	8-1/2
1C-3500-1.50	1-1/2	182TCN			7-1/4	8-1/2
1C-3500-2	2	213TCN			7-1/4	8-1/2
2C-3500-2	2	182TCN			7-1/4	8-1/2
2C-3500-3	3	184TCN			7-1/4	8-1/2
2C-3500-5	5	213TCN			7-1/4	8-1/2
3C-1750-0.33	1/3		56			
3C-1750-0.50	1/2		56			
3C-3500-1	1	182TCN			7-1/4	8-1/2
3C-3500-1.50	1-1/2	182TCN			7-1/4	8-1/2
3C-3500-2	2	182TCN			7-1/4	8-1/2
3C-3500-3	3	184			7-1/4	8-1/2
4C-1750-0.75	3/4		56			
4C-1750-1	1		182TCN			
4C-3500-5	5	213TCN			7-1/4	8-1/2
4C-3500-7.50	7-1/2	215TCN			7-1/4	8-1/2
5C-1750-0.75	3/4		56			
5C-1750-1	1		182TCN		7-1/4	8-1/2
5C-3500-2	2	182TCN			7-1/4	8-1/2
5C-3500-3	3	184TCN			7-1/4	8-1/2
5C-3500-5	5	213TCN			7-1/4	8-1/2
5C-3500-7.50	7-1/2	215TCN			7-1/4	8-1/2
6C-1750-2	2		182TCN		7-1/4	8-1/2
6C-1750-3	3		184TCN		7-1/4	8-1/2
6C-1750-5	5		213TCN		7-1/4	8-1/2

See footnotes at end of table.

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Table II. Ac motor 50°C ambient, class B, F, H, or N insulation with 70°C temperature- rise (see 3.2.14.2). - Continued

Pump	Frame number					
	Hp	3500 r/min	1750 r/min	1150 r/min	AJ 1/	AK 1/
7C-3500-10	10	254TCN			7-1/4	8-1/2
7C-3500-15	15	256TCN			7-1/4	8-1/2
7C-3500-20	20	286TCN			9	10-1/2
8C-3500-10	10	254TCN			7-1/4	8-1/2
8C-3500-15	15	256TCN			7-1/4	8-1/2
8C-3500-20	20	286TCN			9	10-1/2
9C-3500-10	10	254TCN			7-1/4	8-1/2
9C-3500-15	15	256TCN			7-1/4	8-1/2
9C-3500-20	20	286TCN			9	10-1/2
10C-3500-10	10	254TCN			7-1/4	8-1/2
10C-3500-15	15	256TCN			7-1/4	8-1/2
10C-3500-20	20	286TCN			9	10-1/2
11C-1750-7.50	7-1/2		215TCN		7-1/4	8-1/2
11C-1750-10	10		254TCN		7-1/4	8-1/2
11C-1750-15	15		256TCN		7-1/4	8-1/2
12C-1150-5	5			254TCN	7-1/4	8-1/2
12C-1150-7.50	7-1/2			256TCN	7-1/4	8-1/2
12C-1150-10	10			256TCN	7-1/4	8-1/2
13C-1150-15	15			326TCNI 2/	14	16
13C-1150-20	20			326TCNI 2/	14	16
13C-1150-25	25			326TCNI 2/	14	16
14C-3500-7.50	7-1/2	215TCN			7-1/4	8-1/2
15C-3500-5	5	213TCN			7-1/4	8-1/2
15C-3500-7.50	7-1/2	215TCN			7-1/4	8-1/2
16C-1750-15	15		286TCNI 2/		14	16
16C-1750-20	20		286TCNI 2/		14	16
16C-1750-25	25		286TCNI 2/		14	16

1/ AJ and AK dimensions are shown on figure 3.
2/ These frame sizes are larger than those shown in MIL-M-17060 to accommodate the larger bearing, shaft, and mounting dimensions invoked by this specification.

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TABLE III. Dc motors 50°C ambient, class B, F, H, or N insulation (no silicone materials) (see 3.2.14.3).

Pump	Frame number						
	Hp	3500 r/min	2800 r/min	1750 r/min	1150 r/min	AJ 1/	AK 1/
1C-1750-0.33	1/3			56			
1C-3500-0.50	1/2	56					
1C-3500-0.75	3/4	56					
1C-3500-1	1	186NC				7-1/4	8-1/2
1C-3500-1.50	1-1/2	187NC				7-1/4	8-1/2
1C-3500-2	2	216NC				7-1/4	8-1/2
2C-3500-2	2	216NC				7-1/4	8-1/2
2C-3500-3	3	218NC				7-1/4	8-1/2
2C-3500-5	5	254NC				7-1/4	8-1/2
3C-1750-0.33	1/3			56			
3C-1750-0.50	1/2			56			
3C-3500-1	1	186NC				7-1/4	8-1/2
3C-3500-1.50	1-1/2	187NC				7-1/4	8-1/2
3C-3500-2	2	216NC				7-1/4	8-1/2
3C-3500-3	3	218NC				7-1/4	8-1/2
4C-1750-0.75	3/4			186NC			
4C-1750-1	1			187NC			
4C-3500-5	5	254NC				7-1/4	8-1/2
4C-3500-7.50	7-1/2	256NC				7-1/4	8-1/2
5C-1750-0.75	3/4			186NC			
5C-1750-1	1			187NC			
5C-3500-2	2	216NC				7-1/4	8-1/2
5C-3500-3	3	218NC				7-1/4	8-1/2
5C-3500-5	5	254NC				7-1/4	8-1/2
5C-3500-7.50	7-1/2	256NC				7-1/4	8-1/2
6C-1750-2	2			218NC		7-1/4	8-1/2
6C-1750-3	3			254NC		7-1/4	8-1/2
6C-1750-5	5			256NC		7-1/4	8-1/2
7C-3500-10	10	284NC				9	10-1/2
7C-3500-15	15	286NC				9	10-1/2
8C-3500-10	10	284NC				9	10-1/2
8C-3500-15	15	286NC				9	10-1/2

See footnote at end of table.

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TABLE III. Dc motor 50°C ambient, classB, F, H, or N insulation (No silicone materials) (see 3.2.14.3). - Continued

Pump	Hp	Frame number					
		3500 r/min	2800 r/min	1750 r/min	1150 r/min	AJ 1/	AK 1/
9C-3500-10	10	284NC				9	10/1-2
9C-3500-15	15	286NC				9	10/1-2
9C-2800-20	20		326NC			9	10/1-2
10C-3500-10	10	284NC				9	10/1-2
10C-3500-15	15	286NC				9	10/1-2
10C-2800-20	20		326NC			9	10/1-2
11C-1750-7.50	7-1/2			284NC		9	10/1-2
11C-1750-10	10			324NC		9	10/1-2
11C-1750-15	15			326NC		9	10/1-2
12C-1150-5	5				286NC	9	10/1-2
12C-1150-7.50	7-1/2				324NC	9	10/1-2
12C-1150-10	10				326NC	9	10/1-2
13C-1150-15	15				366NCI	14	16
13C-1150-20	20				405NCI	14	16
13C-1150-25	25				444NCI	14	16
14C-3500-7.50	7-1/2	256NC				7-1/4	8-1/2
15C-3500-5	5	254NC				7-1/4	8-1/2
15C-3500-7.50	7-1/2	256NC				7-1/4	8-1/2
16C-1750-15	15			365NCI		14	16
16C-1750-20	20			365NCI		14	16
16C-1750-25	25			366NCI		14	16
17C-2800-20	20		326NC			9	10-1/2

1/ AJ and AK dimensions are shown on figure 5.

3.2.15 Controllers. Motor controllers shall be in accordance with MIL-C-2212 for ac and dc and shall have the characteristics as specified (see 6.2.1).

3.2.16 Painting. External unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment conforming DOD-P-15328 and one coat of zinc chromate primer conforming to TT-P-645, followed by a finish coat of light gray equipment enamel conforming to MIL-E-15090.

3.2.16.1 Painting of external surfaces of nonferrous parts of pumps will not be required.

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3.2:17 Identification plates.

3.2.17.1 Identification plates shall be furnished on each pump and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic or aluminum shall not be used.

3.2.17.2 Identification plates shall be secured to equipment with corrosion-resistant metallic screws or rivets.

3.2.17.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
  (1) Capacity in gallons per minute (gal/rein).
  (2) Total head in lb/in^2.
  (3) Speed of shaft in r/rein.
  (4) Brake hp.
  (5) Test pressure.
  (6) Special data vital to the-unit:
    a. Suction pressure.
    b. Submergence.
    c. Impeller diameter.
(f) Contract number (and item number for multiple unit orders).
(g) National stock number.
(h) Section for Defense Contract Administration Service Managment Area (DCASMA) stamp.
Navy standard pump number.
Coponent identification number (CID) consisting of nine digits that will be supplied to the successful bidder. No materials shall be shipped until the CID is included on the identification plate.
(k) Nameplates to specify actual characteristics of maximum head capacity (WIC) on top half with space provided on botton half to Specify new characteristics as a result of the trimming The following warning instructions shall be stamped in the middle of the nameplate of the pump unit:

"WARNING: The pump included in this pump unit contains a maximum diameter impeller. Before installation, verify' required operating characteristics. Refer to instruction book for impeller trimming instructions. After impeller in this unit is trimmed, complete lower section of nameplate <."

(l) For pumps that cannot accommodate one nameplate containing the complete data required, the use of a standard nameplate on the pumps plus a bottom section nameplate for MHC pumps, mounted directly' below the standard nameplate is acceptable.A middle nameplate (for MHC pumps only) may be a separate nameplate for attaching to the side of the mounting bracket.

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3.2.17.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.2.18 Piping, valves,and threaded items.

3.2.18.1 Unless otherwise specified (see 6.2.1), valve flanges, fittings, and bolting shall conform to MIL-STD-777. All pump flanges shall have a face circular lay finish as detailed in the applicable general note of MIL-STD-777.

3.2.18.2 Tapered pipe threads are acceptable for use in grease fittings located on nonmoving parts. Fittings not covered by MIL-STD-777 for application when ship connections are not involved shall be submitted for review by the design review agency.

3.2.19. Locking device. Internal threaded fasteners and threaded machine elements shall be secured by locking devices where continuous operation under the condicion of Shock, vibration, and temperature specified herein depending" maintenance tight connections of parts. All setscrews internal to the pressure boundary of the pump and all setscrews on internal and external rotating portions of the pump shall be secured by locking devices. Locking devices shall be subject to approval by the design review agency.

3.2.20 Reverse rotation. Pumps shall withstand reverse rotation for a period of 1 minute at all speeds up to design speed without damage.

3.2.21 Provision for handling. Eyebolts, lugs, holes, and other means shall be provided to permit attachment of lifting gear for lifting the assembled pump, driver, and attached accessories as a complete unit. Means shall be provided for the handling of parts and components weighing 150 pounds and over which cannot be handled manually during unit overhauls and preventive maintenance inspections.

3.3 Detail requirements.

3.3.1 Performance characteristics. The performance characteristics of the pump shall be as specified (see 6.2.1).

3.3.2 Detail design characteristics.

3.3.2.1 Pumps for a specific application shall be selected from the rating chart shown on figure 6 for ac motors, or from the rating chart shown on figure 7 for dc motors. The standardized pump size with the hp and motor speed shown on the rating chart for the application shall be furnished. When specified (see 6.2.1), pumps shall be of MHC design. MHC pumps are those furnished with an impeller of the maximum diameter that will not overload the motor at any point along the performance curve.

3.3.2.2 Pumps for all services shall have constantly rising head-capacity curves from maximum or side-open capacity to zero *or* shut-off capacity.

3.3.2.3 Pumps shall be capable of satisfactory operation at a dynamic suction lift of 15 feet of water at 85°F.

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3.3.2.4 Pump units shall not overload the motor at any condition of operation from shut-off capacity to wide-open while pumping the specified fluid.

3.3.2.5 Pumps shall operate satisfactorily at any condition of operation from shut-off capacity to wide-open while pumping-the specified fluid.

3.3.2.6 Pumps shall be right hand rotation (clockwise when viewed from the motor end of unit) only.

3.3.2.7 Pumps shall be fitted with casing wearing rings which shall be secured from rotating by set screws mounted in the joint between the casing and ring.

3.3.3 Casings.

3.3.3.1 Casings shall permit the examination and replacement of wearing parts. Suction covers shall be removable to permit removal of the impeller or casing wearing ring without dismantling the unit.

3.3.3.2 Pumps shall have casings arranged so that the discharge may be located in any of the three positions shown on figure 1.

3.3.3.3 Casings shall be provided with the necessary bosses for vent, drain, priming, and gauge connection in accordance with MS16142 for the thread positions of discharge. Bosses for gauge connections shall be of sufficient size for conversion to flanged connections to socket weld type design suitable for a nipple and flanged connections in accordance with Drawing B-214 (see 6.2.1).

3.3.3.4 Casings shall be provided with a vent cock in the proper location for the position of discharge. The remaining unused vent and drain hole bosses shall be provided with plugs in accordance with MS18229.

3.3.3.5 Casings shall withstand the applicable hydrostatic pressures shown on figure 1 (see 4.6.2).

3.3.3.6 Spare casing wearing rings (see 6.5.2) shall be machined undersize by at least 0.050 inch on the diameter in order to permit finish machining during shipboard installation to mate with used impellers.

3.3.4 Suction and discharge connections. Suction and dicharge connections shall be flanged for mating with flanges in accordance with MIL-STD-777.

3.3.5 Impellers.

3.3.5.1 Impellers shall be of the single inlet closed type. Outside and inside surfaces of the impellers shall be smooth. Impellers shall be dynamically balanced.

3.3.5.2 Impellers shall be keyed on the shaft and securely held against lateral movement by locked nuts or other means acceptable to the design review agency. Shaft threading shall be counter to the direction of rotation.

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3.3.5.3 I-repellers shall not be furnished with wearing rings. The impeller wearing surfaces shall have material thickness to permit reducing the diameter of the impeller wearing surfaces by as much as 0.050 inch to accommodate undersize casing wearing rings to restore design running clearance.

3.3.5.4 The impeller shall have the approximate maximum diameter for each pump as specified in table IV.

TABLE IV. Approximate maximum impeller diameter.

Pump number	Approximate maximum impeller diameter (inches)	Pump number	Approximate maximum impeller diameter (inches)
1C	5	10C	7-1/2
2C	6-3/4	11C	9
3C	5	l2c	9-3/4
4C	6-1/4	13C	10-1/2
5C	5-1/8	14C	8
6C	7	15C	5-1/4
7C	10	16C	9
8C	9-1/4	17C	10-1/2
9C	9-1/4	---	----

3.3.5.5 All rotors shall be dynamically balanced with all rotating parts connected thereto. This requires dynamic balance with the rotating elements of the driving unit in place; however, rotating parts may be balanced individually provided that when assembled the unbalance shall not exceed the limits specified in MIL-STD-167-1.

3.3.6 Shafts.

3.3.6.1 Shaft flingers shall be provided adjacent to stuffing boxes for the pumps.

3.3.6.2 Shafts shall be made of nickel-copper or nickel-copper-aluminum alloy, one-piece construction.

3.3.6.3 Shaft sleeves shall have the dimensions shown on figure 8 (see 3.3.7) and shall suit the mechanical seal.

3.3.6.4 O-ring seals shall be used between the shaft and stuffing box shaft sleeve. Other positive methods of sealing may be used subject to the approval of the design review agency.

3.3.7 Stuffing boxes and mechanical shaft seals.

3.3.7.1 All pumps shall be provided with mechanical shaft seals. Magnetic type seals shall not be used.

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3.3.7.2 Pumps in salt water-service with a total head of 30 lb/in^2 or more shall be fitted with cyclone separators. The separator shall be integrally mounted on the pump (preferably by means of a bracket mounted at the suction head parting flange) and shall be completely connected to the suction and discharge by means of tubing and fittings.

3.3.7.3 Cyclone separators shall be constructed of nickel-copper alloy. Fittings for abrasive separators and pump casings shall be of the straight thread type adapter with O-ring seals and shall be constructed of nickel-copper alloy. Tubing shall be copper-nickel (70-30) in accordance with MIL-T-16420. Multiple springs for sea water service pumps shall be constructed from UNS N06625 (inconel 625) in accordance with ASTM B 446 Or UNS N10002 (Hasteloy C) in accordance with ASTM A 494.

3.3.7.4 Each seal shall be provided with a solid gland plate capable of accommodating two or more rings of packing in accordance with MIL–P-24337 for use in the event of a mechanical seal failure. The gland seal shall also be provided with a flush connection to ensure positive liquid pressure is supplied to the seal faces under all operating conditions and that there is adequate circulation of the liquid at the seal faces to minimize deposits of foreign matter in the seal parts.

3.3.7.5 Metal seal internals shall be nickel-copper alloy or highly alloyed steel in accordance with ASTM A 744, grade CN-7M or CN-7MS. The mechanical shaft seal for the pumps shall be a solid tungsten carbide against a solid carbon face seal.

3.3.7.6 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of set screws.

3.3.7.7 Stuffing box glands shall be set up by nuts threaded on studs secured in the casings or seal gland plate.

3.3.7.8 Pump certification data shall contain certification by the seal manufacturer that there will be adequate circulation of liquid and adequate lubrication of the seal faces when the seal is installed as shown on the sectional assembly drawing (see 3.6.1.2.4.2(m)).

3.3.8 Material. The materials of the pump shall conform to the materials specified in table V. However, this specification is not intended to be restrictive provided proposed alternative materials will give equal or better service than the material specified. Proposed alternative materials shall be subject to approval by the design review agency. Components of the pump for which the specific materials are not specified shall be materials best suited for the intended service. Materials which can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specifications. Particular attention should be given to avoiding sensitization of materials during hard facing, stress-relieving, or repair welding. Use of cadmium-plated parts and fasteners, including washers, is prohibited.

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3.3.8.1 Recovered materials. Unless otherwise specified herein, all equipment, materials and article incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opesed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allow under this specification unless otherwise specified.

TABLE V. Materials

Application	Material	Specification	ASTM publication
Casings	Copper-nickel alloy	MIL-C-20159	B 369, UNS C96400
Studs, plugs, adapters, bolts, nuts and petcocks 1/	Nickel-copper alloy	QQ-K-281, class A or B	B 164, UNS NO4400
	Ni-Cu-A1 alloy	QQ-N-286	---
Impellers	Nickel-copper alloy	W-N-288, comp. E	---
Casing wearing rings 2/	Nickel-copper alloy	QQ-N-288, Comp. B or D	----
		MIL-C-15345, alloy 16 or 17	---
Shafts 3/	Nickel-copper alloy	QQ-N-281, class A or B	B 164, UNS N04400 or N04405
	Ni-Cu-A1 alloy	QQ-N-286, class A or B	
Shaft sleeves, stub or step	Ni-Cu-A1 alloy	QQ-N-286, class A or B, min. Bhn 265	---
	Nickel-copper alloy	QQ-N-281, class A or B	B 164, UNS N04400 or N04405
Flingers and mounting brackets	Gun metal	QQ-C-390, alloy C90300	B 584, alloy C90300 or C90500
	Copper-nickel alloy	MIL-C-20159, type I	B 369, alloy 964 UNS C96400
	Valve bronze	QQ-C-390, alloy C92200	B584, alloy C92200 or B 61

See footnotes at end of table.

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TABLE V. Materials. - Continued

Application	Material	Specification	ASTM publication
Impeller nuts, washers, and keys	Nickel-copper alloy	QQ-N-281, class A or B	B 164, UNS N04400 or N04405
Casing gaskets	Asbestos, sheet, compressed	HH-P-46	------ ------ ----
Seal internals	Nickel-copper alloy	w-N-288	---- -- - -- - -- - - - -
	Highly-alloyed corrosion resist-ing steel	------ ------ ---	A 744, grade CN-7M or CN-7MS
Cyclone separ-ators	Nickel-copper alloy	W-N-281 QQ-N-288	-- - --- - - -- ----
Tubing and nipple	Copper-nickel alloy	MIL-T-16420 (70-30) class 700	-- ---- - - - -- - - - - -
Fittings, silver brazing O-ring face seal type 1/	Nickel-copper-alloy	QQ-N-281, class A or B	B 164, UNS N04400 or N04405
	Copper-nickel alloy (70-30)	MIL-C-15726 (70-30)	-- - -- -- ---- - - --
Flanges (gauge and vent connections)	Copper-nickel alloy (70-30)	MIL-C-15726 (70-30)	-- -- -- - - - - - --- --

1/ Mechanical flareless (bite), compression, or flared type fittings are not permitted.

2/ Wearing rings and opposing surfaces shall be of dissimlar material or shall have a difference in hardness of at least 50 Brinell hardness number (Bhn).

3/ The Ni-Cu-A1 material shall be annealed prior to age hardening.

3.3.9 Part interchangeability. All parts for pumps manufactured to the same drawings shall be interchangeable.

3.3.9.1 Pumps of a given capacity In gal/rein and total dynamic head in feet and type of electrical voltage shall be interchangeable in mounting dimensions, size and location of piping connections, shaft extension, shaft sleeve, and flinger.

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3.4 Tools.

3.4.1 Tools which are available in the Federal Supply Catalog, such as common wrenches and standard pullers and extractors, will not be required. Standard complete or limited sets of pullers or extractors or parts of puller sets in accordance with GGG-P-781 or GGG-E-950 shall be identified as to type and use in notes on the outline or sectional assembly drawings. Special tools shall be detailed and included in the list of material. Special tools required for the maintenance and repair of the pump units shall be furnished The number of sets of special tools of the pump units shall be as specified (see 6.2.1). Special wrenches shall be of forged steel with hardened jaws. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of the catalog may be consulted in the office of the DCASMA.)

3.5 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2 .2) and as specified in 3.6 through 3.8.2.

3.6 Drawings. When required (see 6.2.1), the contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2) and the unique features specified in 3.6.1 through 3.6.4 shall be included.

3.6.1 Drawing content.

3.6.1.1 General requirements.

3.6.1.1.1 "Manufacturer's use only" notes. Information intended for manufacturer's use only shall be so designated. Conversely, on a commercial drawing with wide usage it is permissible to designate portions thereon, "For Navy use only".

3.6.1.1.2 Dimensional tolerance. Unless otherwise indicated by note, legend, or key all tolerances shown on a drawing must be complied with during both manufacture and maintenance repair.

3.6.1.2 Design drawing. Design drawings shall consist of the drawings described in 3.6.1 .2.1 through 3.6.1.2.4.

3.6.1.2.1 Sectional assembly drawing. Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts. All running clearances shall be shown and shall be dimensioned as radial clearances. (The radial clearance equals one half the diametral clearance.) This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view. Tighten torques and thread lubrication requirements for threaded fasteners shall be shown on the assembly drawing.

3.6.1.2.2 List of materials. The list of materials shall include every part required in the pump assembly, including those parts not required to be detailed. The list of material shall include an indication of each part required by this specification to be furnished as an on board repair Part (this is not a repair parts list). The list of material shall include the manufacturer's drawing number and service part number for all parts for which

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detail drawings are required (see .3.6.l.2.3) as well as quantity required, material type, and materials specification for all parts.Parts shall be named to indicate the function they serve.Parts for which detail drawings are not required shall be identified as the dimensions. The list of materials may be shown on the sectional assembly drawing.

3.6.1.2.3 Detail drawing. Detail drawings shall be furnished of all parts and sub-assemblies necessary for evaluation of the equipment and of all parts necessary for maintenance and overhaul of the pumps. Detal of these parts shall be-so complete as to permit emergency manufacture by a Naval ship-yard without assistance from the original manufacturer. Details of pump casing shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Sub-assemblies whose parts cannot be acquired or serviced individually shall be shown as a single part and so indicated. Multi-detail drawings are preferred, but monodetail drawings may be used.

3.6.1.2.3.1 Impeller machining and inspection drawing. Detail drawings shall explicitly require that each impeller, including spare and replacement impellers, be dynamically balanced to less than the maximum ounce-inch residual imbalance allowed by MIL-STD-167-1 (type II) for the impeller's weight and rotational speed.

3.6.1.2.4 Outline drawings and certification data.

3.6.1.2.4.1 Outline drawings and certification data shall be in the form of supplemental drawings to all sectional assembly and detail drawings. Separate outline drawings, drawing lists, and certification data shall be furnished under each contract or order unless the complete equipments covered by the outline drawing and the referenced drawings are in fact identical in all respects.

3.6.1.2.4.2 The outline drawings, in addition to the certification data required, shall contain the following:

(a) Dimensional outline assembly drawing of the pump with its prime mover, bedplate, and attached auxiliaries.
(b) Complete performance data of pump, prime mover, and attached auxiliaries , if applicable.
(c) Complete equipment performance curves, based on actual tests (the original submission may show design performance curves vice test curves).
(d) Weights of major sub-assemblies over 50 pounds, such as, rotor, upper half-casing, and impeller; and the weight of the complete unit including driver, dry and in operating condition.
(e) Shipbuilder's connections showing size, type, and dimensions of flanges.
(f) Center of gravity of the pump component and of the complete assembly.
(g) Radii of gyration of complete assembly about each of the three principal axes.
(h) Identification of system in which installed.
(i) Critical speed.
(j) Maximum thrust loads.

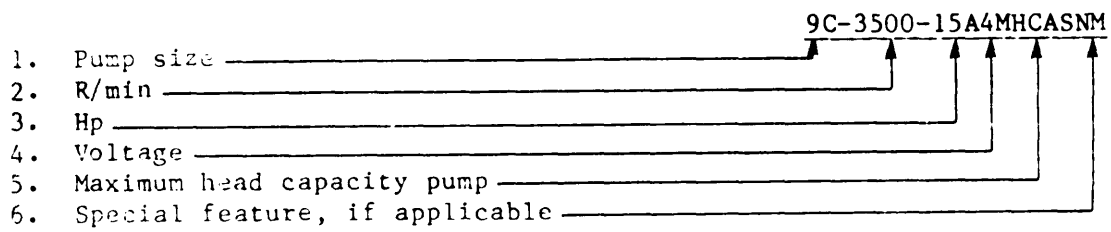
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(k) - Endurance tests, high-impact shock tests, and non-destructive tests (if performed) and the design review agency letters approving these reports or extension of any tests.
(l) Overall dimensions of the complete unit including locating of all fittings and connections, the space required for removal and replacement of parts for maintenance, and mounting information.
(m) The certification by the mechanical seal manufacturer required by 3.3.7.8.

3.6.1.2.4.3 The pump performance chart included on the certification data, outline drawing, or both shall include performance curves depicting gal/rein, feet head, efficiency, and brake hp for various impeller diameters to provide Navy field activities with impeller trim data. Sufficient performance curves shall be shown to provide impeller trim data for head capacities included within the total rating boundary for the pump show on figures 6 and 7.

3.6.1.2.4.4 All drawing shall include in the title block the designation "Navy Standard Close-Coupled Pump, SPCC Identification_____."

NOTE: This identification is developed from the following sample symbol system:



Permissible symbols are as follows:

1. Pump size - as required by contract
2. R/rein - as required by contract
3. Hp - as required by contract
4. Voltage - A = ac
   d = dc
   1 = 115-volt
   2 = 230-volt
   4 = 440-volt
5. Maximum head capacity pump - always MHC
6. Special feature, if applicable - AS - abrasive separator
   NM - nonmagnetic

3.6.1.2.4.5 Drawing content for electric motors shall be in accordance with MIL-M-17060 for Navy service A.

3.6.1.2.5 On board repair parts list. A list of on board repair parts shall accompany the outline drawings. This list shall be provisioning list forms in accordance with MIL-P-15137. This list shall be furnished whether or not the parts are required in the pump contract or order. This list shall not be modified to indicate parts or quantities of parts

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furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and purchaser at the time of equipment drawing approval. The list of on board repair parts actually purchased shall be prepared as required by MIL-P-15137 (see 6.5.2).

3.6.1.2.6 Review. Design drawings shall be submitted to the design review agency for approval (see 6.4).

3.6.1.3 On one of the delivered drawings, the mechanical seal vendor shall identify and certify the following:

(a) Whether or not there is an adhesive or other setting compound factory-installed on the elastomer bellows or O-ring that seals between the mechanical seal and the impeller hub or shaft sleeve.
(b) The description of any such adhesive or setting compound.
(c) Any time limit associated with that compound when installed on a mechanical seal in stowage and any time limit between initial wetting of that compound and the final positioning of the mechanical seal in the pump.
(d) The type and amount of lubricant, if any, to be used to facilitate axial positioning of the mechanical seal in the pump during seal installation. The drawing shall require the manufacturer to use that lubricant during pump assembly.

3.6.2 Material identification.

3.6.2.1 Preferred material reference. Where materials of identical or equal quality can conform to more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced, the following is the order of preference:

(a) Industry and technical society specification or standard.
(b) Federal specification or standard.
(c) Military specification or standard.
(d) Manufacturer’s specification or standard.

3.6.2.2 Material substitions. Where materials other than those covered by (a), (b), and (c) of 3.6.2.2.1 are referenced and approved by the design review agency, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the q aterial in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary from Navy materials stocks.

3.6.3 Drawing identification. Prime vendors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the parts delineated thereon are produced by the subcontractor. The prime contractor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

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3.6.3.1 Prime contractors who purchase semi-finished parts from sub-contractors for final production, test, or selection phases in their own shops have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.6.4 Drawings of drivers and associated equipment shall be in accordance with the specifications covering those equipments.

3.7 Technical manuals.

3.7.1 Unless otherwise specified in the contract or order (see 6.2.1), technical manuals shall be prepared in accordance with type I of MIL-M-15071. The modifications and clarifications specified in 3.7.2 through 3.7.5 shall apply. A separate manual shall be furnished for each different pump unit except in special cases When manuals covering more than one pump may be approved . Preliminary manual shall be submittedto the design review agency for appoval and shall include all proposed section complete. Unless otherwise specified in the contract or order, manuals shall include pump, driver, controls, and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

3.7.2 Manuals covering pumps driven by conventional electric motors with conventional controllers need need include operation and maintenance instructions covering the motors and controllers . They shall , however, include master drawings and certification data covering the motors and controllers. Complete operation and maintenance instructions covering electrical equipment shall be included when specifically required (see 6.2.1).

3.7.3 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawing.
(b) Outline drawing.
(c) Complete list of material corresponding to the sectional assembly drawing.
(d) Certification data.
(e) A minimum of two photographs of the complete unit, pump and driver , taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied, photographs of one design unit only need be supplied.

3.7.4 The alignment procedure instructions in the manual shall describe in complete detail the means by which the required alignment clearance at sleeve bearings and wearing rings is to be established. The manual shall prescribe the maximum permissible eccentricities which can be tolerated in the aforementioned areas with the pump in the assembled condition. Exceptions to any part of the aforementioned requirements in any specific area may be had only by providing for the design review agency approval a written technical justification for the said exception. The manual shall specify the maximum allowable clearance for wearing rings and sleeve bearings before replacement of the parts. In addition, the manuals shall contain:

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(a) Lubrication schedule including lubricant and procedure.
(b) A statement directing the user of the technical manual to see the Centrifugal Pump Chapter of the Naval Ships Technical Manual for the criteria for replacing mechanical seals because of leakage.
(c) Torque values and sequence required to maintain proper seal and alignment.
(d) In-field dynamic balancing instructions where provisions for balancing in field are required.
(e) The mechanical seal installation procedure including complete information regarding adhesive, the adhesive time limits and lubricant (see 3.6.1.3), and any additional guidance needed to avoid causing an excessive preload on the mechanical seal.

3.7.5 The quantity and distribution of manuals shall be as specified (see 6.2.1).

3.8 Technical repair standard. A technical repair standard (TRS) shall be provided upon initial introduction of a new equipment or in-production of an equipment that has an approved TRS if major or minor configuration changes have been approved that would effect the adequacy of the standard for use in new equipment repairs (see 6.2.2).

3.8.1 TRS technical content shall include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment's dimensions, clearances, and tolerances such that the equipment is capable of performing its function as originally specified and is capable of being logistically supported by the DoD logistics support system.

3.8.2 The TRS format and content shall conform to DOD-STD-2147 and shall be approved and validated in accordance with this standard.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for comliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Inspection system program plan. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.1.3 Variations. No variations or deviations from approved design drawings in 3.6.1.2 will be permitted.

4.1.4 Certificate of compliance with mercury prohibition. When specified (see 6.2.1), pumps, drivers, and auxiliary parts furnished by the pump contractor shall be free of mercury contamination and free of functional mercury. During the manufacturing processes, checks, examinations and tests, the product being offered for acceptance shall not come in direct contact with mercury, any of its compounds, nor with any mercury containing device, such as gauges and thermometers. The manufacturer shall certify that the product, when shipped, is free from, mercury contamination (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article tests.First article inspection shall be performed at a laboratory satisfactory to NAVSEA on each class, Either horizontal or vertical units may be submitted. The several first article tests on one pump design may be conducted concurrently if practicable. The following tests shall be performed:

(a) Test procedures and test reports (see 4.6.1).
(b) Hydrostatic test (see 4.6.2).
(c) Operational test (see 4.6.3).
(d) Performance tests (see 4.6.4).
(e) Mechanical shaft seal test (see 4.6.5).
(f) Shock test (see 4.6.6).
(g) Vibration test (see 4.6.7).
(h) Endurance test (see 4.6.8).
(i) Airborne and structureborne noise tests (see 4.6.9).

4.3.1 Motors shall be tested in accordance with the applicable motor specification prior to assembly with the pump.

4.3.2 First article shall include all sizes for either horizontal or vertical units.

4.4 Quality conformance inspection. Each pump (and subcomponents of each pump, if applicable) shall be examined and tested as specified below. Quality conformance inspection shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test (see 6.2.2). A compilation of the documentation of all the quality conformance tests performed on each pump shall be furnished (see 6.2.2).

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(a) Hydrostatic test (see 4.6.2)
(b) Operational test (see 4.6.3).

4.4.1 In addition, one pump of each size on the contract or order shall be subjected to the following tests:

(a) Performance tests (see 4.6.4).
(b) Mechanical shaft seal test (see 4.6.5).
(c) Shock test (see 4.6.6).

4.5 Tests. The contracting activity and Government representatives shall have the right to examine the facilities at the contractor's plants and at the contractor's subcontractor's plants, and to witness all tests specified herein. Failures, deficiencies, and discrepancies revealed during the performance of the specified tests and the corrective measures taken shall be recorded and fully documented in the applicable test records and test reports (see 6.2.2). After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability for the modified punp. Major failure indicative of the design deficiency (as distinguished from shop error or faulty workmanship) shall be reported to the design review agency before a correction is made.

4.6 Tests.

4.6.1 Test procedures and test reports. The contractor shall prepare test procedures and test reports in accordance with the data ordering documents included in the contract or order (see 6.2.2). All first article test reports shall be submitted to NAVSEA for approval.

4.6.1 Test procedures and test report. Each pump casing shall be tested hydrostatically to pressures shown on figure 1. Hydrostatic test pressures shall be maintained for at least 15 minutes, or longer as necessary for examination of the entire casing, The casing and suction cover shall exhibit no leakage through the casing material or joints.

4.6.3 Operational test. Each pump assembled with its driving motor shall be tested at its rated operating speed and point of rating by a continuous non-stop run of at least 30 minutes, to check operation and smoothness of running.

4.6.4 Performance tests.

4.6.4.1 Performance tests shall be conducted and reported in accordance with the Test Code of Standards of the Hydraulic Institute, Centrifugal Pump Section, for tests classed as *'Factory tests at the pump manufacturer's plant'".

4.6.4.2 Driving units shall be tested as required by the applicable motor specifications (see 6.2.1).

4.6.4.3 One pump of each size on the contract or order, complete, including the motor, shall be given a test to determine the overall power consumption of the unit under the specified or guaranteed conditions. The command *or* agency concerned may consider waiver of overall power consumption tests on pumps if identical units with identical guarantees have been previously manufactured and tested by the same contractor.

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4.6.4.4 Performance tests shall adequately demonstrate the ability of the pump to handle its rated capacity of specified liquid at the maximum temperature and minimum suction head, or maximum suction lift or varuum., as applicable. For MHC pump, these tests shall be performed with the maximum diameter impeller allowed for the size of the pump being tested.

4.6.4.5 Sufficient data shall be taken during the tests to prepare pump characteristic curves as specified in 4.6.4.6. The test data shall be corrected to the specified operating conditions as to voltage, frequency, temperatures, specific gravity, suction head or lift, and vacuum, as specified in the contract or order; such conditions shall be clearly shown on the data sheets. The data sheets shall state the actual finished diameter of the pump impellers installed on test. Test data and curves shall be complete over the entire range of capacities from shut-off to as near free delivery as possible.

4.6.4.6 The following test curves shall be supplied for the rated speed conditions of the motor. Head capacity and brake hp curves shall demonstrate capacities from over-loaded at any point from shut-off to wide open.

(a) Total head versus capacity.
(b) Pump efficiency versus capacity.
(c) Brake hp versus capacity.
(d) Life versus capacity.

4.6.4.7 Test reports for the performance evaluation tests (see 4.6.4.1) shall be compiled into a single document. The document shall identify the pump by manufacturer’s serial numbers of pump and driver, the contract or order number under which the tests were performed, the ship or class of ships for which the pump is intended if known, and the letter and date of the test procedure acceptance. The test compilation document shall be distributed as specified in 6.2.2.

4.6.4.8 Unless previously performed, one pump of each size shall also be given additional tests using dynamometer or calibrated motor (motor manufacturer’s efficiency test may be used to identify motor calibration) to produce the following curves:

(a) Gal/rein versus total dynamic head at constant rated r/rein for various impeller diameters with superimposed constant efficiency and constant brake hp curves at 1.00 specific gravity. The number of impeller diameters shall be sufficient to develop a family of curves.
(b) Determination of the maximum suction lift in feet versus gal/rein at the maximum impeller diameter tested.

4.6.4.8.1 Two sets of data and curves based on the additional tests, including description of the test set-up, shall be forwarded directly to the acquisition agency, via the DCASMA, and shall not be included as a part of the regular performance acceptance tests.

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4.6.5 Mechanical seal and alignment test. The mechanical seal and alignment test shall be as follows:

(a) The mechanical seal on one pump of each size on the contract or order shall be installed and aligned precisely in accordance with the procedures stated in the proposed pump technical manual. Packing shall not be installed during this test. The amount of force required to axially position the mechanical seal onto the impeller hub, the shaft sleeve, or whatever it rests on in the assembled condition shall be measured and recorded. If the mechanical seal includes an adhesive or setting compound on the elastomer that seals between the mechanical seal and the impeller hub or shaft sleeve, and if there is a maximum allowed time between initial wetting of that compound and final positioning of the mechanical seal in the pump, then do not perform that final positioning until the maximum allowed time has clasped.

(b) After installation, the mechanical seal shall be run in and the pump operated for at least 4 hours at maximum rated speed with the pumped fluid at maximum operating temperature and with pump suction pressure as high as feasible without requiring pump test loop modifications. The seal leak rate shall then be measured and recorded. Also, the dynamic balance shall be measured and recorded using a vibration indicator of the neon tube or similar type capable of measuring vibrations of 0.001 inch or less. During subsequent pump disassembly, the final axial position of the mechanical seal shall be measured and recorded if the pump design permits.

Acceptance criteria: Mechanical seal leakage shall not exceed 5 drops per minute. The pump and drive shall exhibit no abnormal noises or roughness of operation. Vibration shall not exceed the limits specified in MIL-STD-167-1, type II, measured on the bearing caps. The mechanical seal shall have been positioned at the axial position recommended by the mechanical seal vendor to generate the proper amount of mechanical seal spring force. The mechanical seal vendor shall provide in writing his recommendation regarding the axial position and his concurrence that the mechanical seal alignment test results demonstrated proper positioning of the seal. If it was not practicable to measure the axial position of the mechanical seal, then the seal vendor shall concur that the recorded force used to position the seal during assembly was acceptable and shall state the technical basis for considering that force acceptable. A copy of these mechanical seal vendor documents shall be provided as part of the test report.

4.6.5.1 Packing and alignment test. The mechanical seal shall not be installed during this test. The pump shall be packed and aligned precisely in accordance with the procedures stated in the proposed technical manual. The packing shall then be run in the pump and operated for at least 4 hours at a maximum rated speed with the pumped fluid at maximum operating temperature and with pump suction pressure as high as feasible without requiring pump test loop modifications. The time of day of each packing leakage measurement and of each, packing adjustment shall be recorded. The dynamic balance shall be checked with a vibration indicator. After the tests, the packing and shaft sleeve shall be inspected and the results recorded. If the rated maximum operating

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temperature exceeds 200°F and the normal operating temperature is less than or equal to 200°F, the packing and alignment test shall be repeated satisfactorily with the pumped fluid at normal operating temperature'.

Acceptance criteria: The stabilized packing leak rate, defined as that measured at least 30 minutes after the last packing adjustment, shall be within the limit recommended by the pump vendor. Vibration shall not exceed the limits specified in MIL-STD-167-1, type II, measured on the bearing caps. There shall be no smoking or scorching of packing and no scoring (vice polishing) of the shaft sleeve.

4.6.6 Shock test.

4.6.6.1 Unless otherwise specified in the contract or order (see 6.2.1), one complete pumping unit of each size shall be shock tested to assure compliance with the requirements specified in 3.2.3.

4.6.6.2 Equipment exactly duplicating that previously shock tested and accepted need not be retested except when evidence of low shock resistance develops in installed units. NAVSEA will initiate action with the manufacturer for correction of deficiencies and reserves the right to require additional shock tests.

4.6.6.3 Pump units shall be shock tested with drivers. Motors shall be shock tested in accordance with the applicable motor specifications (see 6.2.1).

4.6.6.4 Pump units shall be shock tested running and at standstill on alternate blows.

4.6.6.5 The pump unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of parts, including mounting bolts.
(b) Appreciable distortion or dislocation of a part, such as shaft, mounting feet and bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to test at rated speed, or q ore than that permitted by the specifications for the driver. The amplitude of unbalance both before and after the shock tests and the speed at which measured shall be recorded in the shock test report.

4.6.6.6 If a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory or Government laboratory. Where shock tests are conducted at a Government laboratory, copies of the applicable drawings shall accompany the units.

4.6.6.7 Pump units which have passed the shock test shall be considered acceptable for such service as NAVSEA may authorize. The equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) Bearings shall be replaced.

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(d) Each part shall be carefully examined by the manufacturer and any part which he considers substandard shall be replaced.

4.6.6.7.1 Minor deformations shall be defined as those which do not cause rejection of the design under shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.6.6.8 Reinspection. After shock testing, tests shall be performed verifying performance characteristics and pump integrity.

4.6.7 Vibration test. When specified (see 6.2.1), pumps shall he tested in accordance with MIL-STD-167-1, type II.

4.6.8 Endurance test. The pump furnished for initial qualification shall be operated for a period of not less than 500 hours of actual running time with a minimum of 100 starts to ascertain reliability of performance and operation.

4.6.8.1 Prior to commencement of the endurance test and immediately after completion of the endurance test, the pump shall be disassembled to the extent necessary and the critical dimensions and running clearance of parts subject to wear, erosion, and derangement shall be measured, calculated, and recorded. Components such as pump impellers and casings subject to erosion, corrosion, cavitation, and wear, the effects of which are not subject to routine measurement, shall be listed in the inspection record and after completion of the test, the condition of each component determined and recorded.

4.6.8.2 During the initial and final hours of the endurance test run, the capacity test in accordance with 4.6.4.5 shall be performed to determine the changes in the pump's performance characteristics

4.6.8.3 The endurance test shall not be continuous but shall be interrupted by at least three rest periods of approximately 8 hours each. The number of starts specified in 4.6.8 at full line voltage shall be performed during the course of the test. During an early part of the endurance test the pump shall be operated continuously for 24 hours at a capacity as near free delivery as possible at maximum rated speed and normal specified temperature, submergence and suction conditions. During the latter part of the endurance test the pump shall be operated as near shut-off as possible for 12 hours continuously. The remainder of the endurance test shall be run at maximum rated speed and within plus 20°F, minus O°F of maximum specified liquid temperature. The pump shall be operated at one third, two thirds, and rated capacity in approximately equal time intervals. Operations shall be at minimum specified net positive suction head available, or maximum specified suction lift or vacuum, as applicable.

4.6.8.4 The pump shall be monitored during the endurance test to accurately record the conditions of operation, speed at which operated, and the general performance observed. Data shall be collected and the pump examined at least twice per day of operation. For each periodic examination, in addition to all measured data, the record shall indicate the following:

(a) The pump seal leakage rate.
(b) The smoothness of operation (normal-abnormal).
(c) Other abnormal findings.

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(d) Adjustments made.
(e) Changes made in the conditions or method of operation.

4.6.8.5 Endurance test acceptance criteria:

(a) The head-capacity curve at maximum rated speed after 5OO hours of pump operation shall conform to the specification requirements and shall show no abnormal deviations from the curve before the 500-hour test.
(b) The unit performance and operation after 500 hours of operation shall be unchanged and normal and meet all specification requirements.
(c) The unit operation at the end of the endurance test shall be smooth and in conformance with the specification.
(d) The leakage rate from a mechanical seal shall not exceed 5 drops per minute..
(e) Running clearances shall be normal.
(f) Components subject to attack from corrosion, erosion, and cavitation shall be in a condition commensurate with 500 hours of service.
(g) Wear rates for wearing parts critical for proper operation shall shall a rate of wear for the test period that shall be consistent with the specified design life requirements.
(h) No failures shall have occurred throughout the 500-hour test.

4.6.8.6 Post endurance test procedure. The unit subjected to the 500-hour endurance test shall be restored to the as–new condition by replacement of all parts worn beyond the as-new design tolerances. The restored unit shall pass the tests specified in 4.6.2, 4.5.3, and, if applicable, 4.6.9 The test documentation shall indicate that the unit was subjected to the endurance test and subsequently restored and tested and that it shall be certified as fully conforming to the specification for unrestricted service.

4.6.9 Airborne and structureborne noise tests. When specified (see 6.2.1), pumps shall undergo an airborne noise test—conducted and reported in accordance with MIL-STD-740. The noise test stand details, instrumentation, and testing techniques shall be submitted by the contractor for review by the contracting activity prior to the test commencement. Noise tests shall be performed with the driver furnished with the pump.

Acceptance criteria:

(a) Airborne noise. The unit shall meet the airborne noise level limits specified. If the first three production units of the contract or order are 10 decibels (dB) lower in all octave bands than those specified, and providing all production units comply with the balance requirements of MIL-STD-167-1, type II or the structureborne noise levels specified in the contract or order (see 6.2.1), then airborne noise testing need not be conducted on the remaining units of the contract or order.
(b) Structureborne noise. When required, the unit shall meet the structureborne noise level limits specified in the contract or order (see 6.2.1). If specific structureborne noise levels are not specified in the contract or order (see 6.2.1), then type II II of MIL-STD-740 shall apply.

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4.7 Packageing inspection. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall -be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Technical data. Unless otherwise specified (see 6.2.1), technical data depicting the preservation, packaging, and transportation of support data shall be prepared by the contractor and submitted to the acquisition activity (see 6.2.2).

5.2 Preservation, packing and marking. Pump units and accessories shall be preserved level A or C; packed level A, B, or C as specified (see 6.2.1); and marked in accordance with MIL-P-16789.

6. NOTES

6.1 Intended use. Pumps covered by this specification are intended for miscellaneous services on board Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Designation of manufacturer (pump or driver) responsible to the purchaser for compatibility of pump and driver and operation of the assembled unit.
(c) Pump number (see figures 6 and 7).
(d) Size and weight requirements (see 3.2).
(e) Performance characteristics (see 3.2 and 3.3.1).
    (1) Rated capacity - gal/rein or MHC.
    (2) Rated total dynamic head - lb/in^2 or MHC.
    (3) Liquid to be pumped.
    (4) Specific gravity of liquid to be pumped.
    (5) Temperature of liquid to be pumped.
    (6) Suction head.
(f) Vibration test requirements (see 3.2.3.4 and 4.6.7).
(g) Noise level limits and whether noise test is required (see 3.2.5 and 4.6.9).
(h) Mounting of units (see 3.2.6.1).
(i) If threaded fasteners are other than specified (see 3.2.10.2).
(j) Piping, flanged connections, and valve requirements (see 3.2* 11.1 and 3.2.18.1).

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(k). If mercury is prohibited (see 3.2.13 and 4.1.4).
(l) Voltage for dc motors (see 3.2.14.3.1).
(m) Ordering data for electrical controllers, if required (see 6.2.1 of MIL-C-2212) including variable speed controllers (see 3.2.15).
(n) Whether MHC design is applicable (see 3.3.2.1).
(o) Whether gauge connections are required (see 3.3.3.3).
(p) Number of sets of special tools required (see 3.4.1).
(q) Drawings required (see 3.6).
(r) Technical manuals required (see 3.7.1).
(s) Whether complete information for electrical equipment is required in technical manuals (see 3.7.2).
(t) Quantity and distribution of technical manuals (see 3.7.5).
(u) Tests of driving units (see 4.6.4.2 and 4.6.6.3).
(v) Shock test requirements (see 4.6.6.1).
(w) Whether packaging data is required (see 5.1).
(x) Levels of preservation and packing (see 5.2).
(y) Identification of contracting activity and design review agency (see 6.4).

6.2.2 Data requirement this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410–6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option	Deliverable to
3.2.13 and 4.1.4	Certificate of compliance	DI-E-2121	------	1,2,3,4,5,6,7,
3.6	Drawings, engineering, and associated lists	DI-E-7031	Level 2 or 3 Design activity designation - Contractor Drawing no. - Contractor Delivery of hard copy - contracting activity	1 ,2,3,4,5,6,7

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Paragraph no.	Data requirement title	Applicable DID no.	Option	Deliverable to
3.8	Technical/maintenance overhaul and repair standards (TRSS)	UDI-L-26481	----	1,2,3,4,5,6,7,8
4.1.2	Inspection system program plan	DI-R-4803	----	
4.4 and 4.5	Procedures, test	UDI-T-23732	----	1 ,2,3,4,5,6,7
4.4, 4.5, 4.6.1 and 4.6.4.7	Reports, test	DI-T-2072	10.1.b	1,2,3,4,5,6,7
5.1	Preservation and packing data	DI-L-7135	--- -	1,2,3,4,5,6,7
6.5	Supplementary provisioning technical documentation	DI-V-7000	----	
	Manufacturer's commercial manual (provisioning)	DI-V-7001	----	
	Provisioning parts list	DI-V-7002	----	
	Short form provisioning parts list	DI-V-7003	----	
	Interim support items list	DI-V-7006	----	1,2,3,4,5,6,7,8

Code no.	Activity
	Naval Sea Systems Command Washington, DC 20362
1	56Y31 (2 copies)
2	08L (Submarines and Nuclear ships only)
3	Acquisition Activity
4	Design Review Agency
5	Cognizant Supervisor of Shipbuilding
6	Shipbuilder
7	Defense Contract Administration Services Management Area
8	Spare Parts Control Center, Mechanicsburg, PA

NOTE : Numbers 3 and 6 may be the same activity and only one COPY thus may be required.

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with. specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item Should be a first article sample. The first article should consist of one unit The contracting officer should include specific instructions in acquisition documents regarding. arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is currently appropriate for the pending contract.

6.4 Definitions. As used herein, “contracting activity” refers to the activity purchasing the equipnent covered by this specification. The contract–ing activity may be the Government or a shipbuilder. The “design review agency” refers to the activity responsible for controlling the technical requirements for dpsign and testing of the equipment. The design review agency is generally a Government command or agency such as NAVSEA or an authorized representative. Communication with the design review agency should be handled through the contracting activity (see 6.2.1).

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5.2 On board. The repair parts listed are shown for guidance only in preparing provisioning technical documentation.

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TABLE VI. On board repair parts.

Name of part	Quantity per set
Ball bearings Felt wiper rings or oil seals Packing, if special Mechanical seal Shaft sleeves	100 percent complete replacement for all installed units per ship.
Undersize casing wearing rings Stuffing box throat bushings Impeller nuts	100 percent replacement for one pump per ship.

6.5.3 Each box containing on board repair parts should contain a list entitled “List of on board repair parts”. The list should be-in a format for use on outline drawings and in the technical manuals. It should be not less than nominal 8-1/2 by 11 inches in size.

6.5.3.1 The format of the list should include a heading and columns of data for the items listed. The heading should include titles and applicable entries as follows:

Reproduced from drawing number
Number of ships
Application
Contract number
Manufacturer
Quantities are for units per ship
The columns should include:
- Piece number
- Name of part
- Quantity
- Drawing number
- Federal stock number

Additional columns may be used as applicable.

6.5.3.2 The list should be so treated as to be resistant to oil, water, and fading.

6.5.4 The repair parts requirements for the drivers and accessories should be in accordance with the related equipment specification.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4320-N255)

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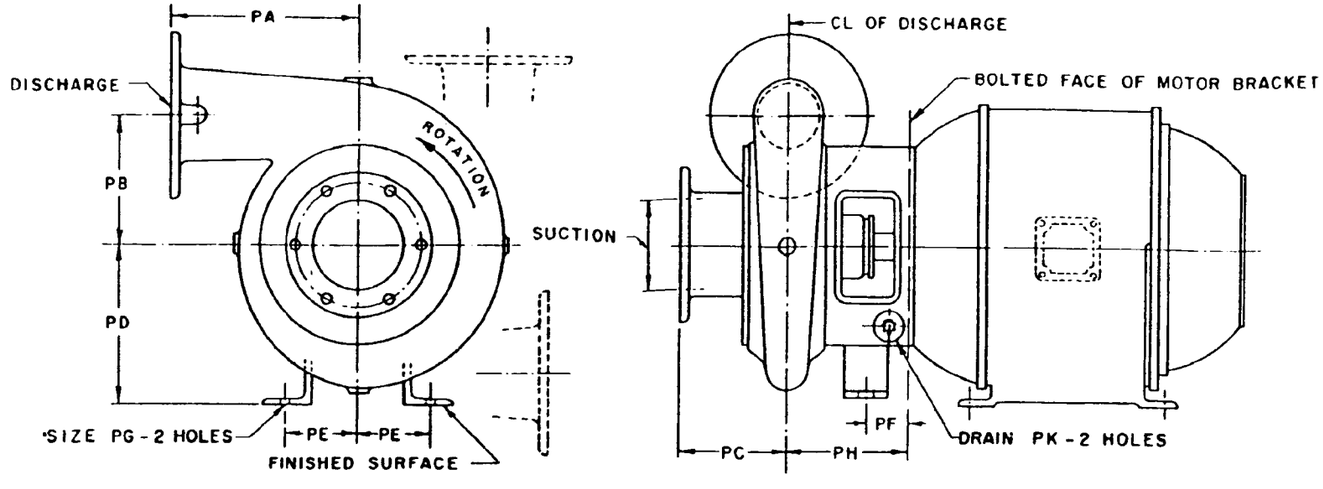
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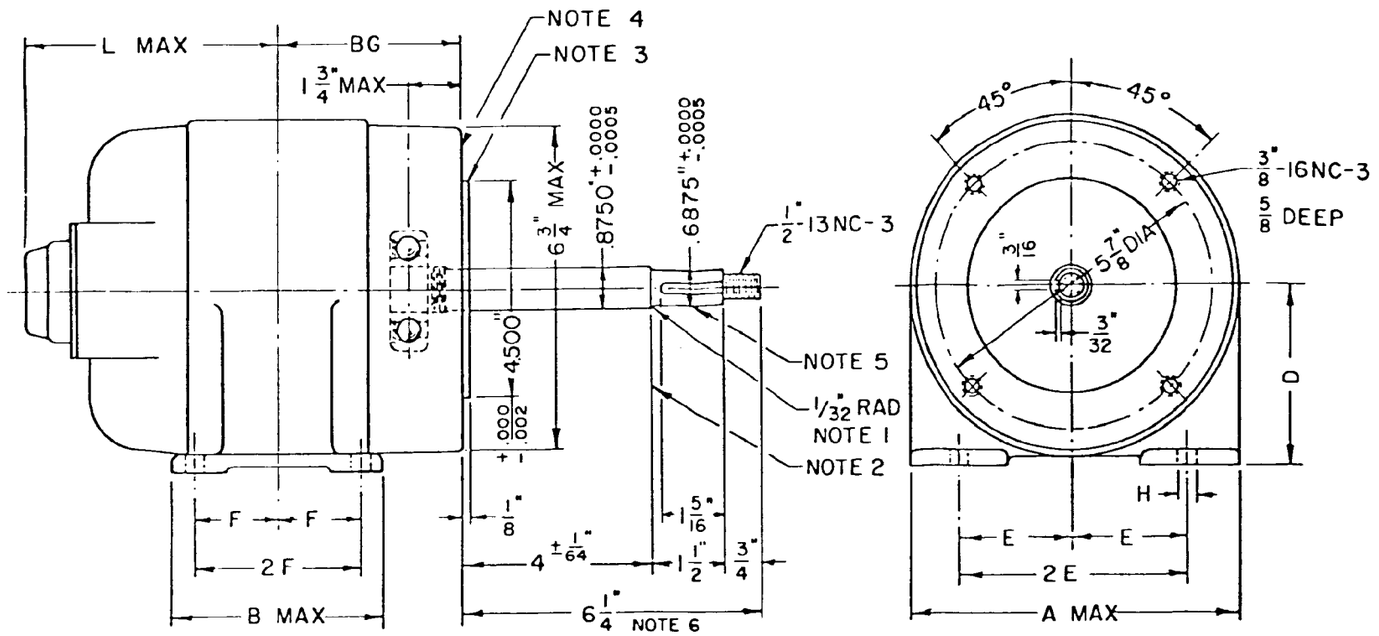
SH 2989A

Pump No.	Navy size flange Disch.	Navy size flange Suction	PA	PB	PC	Hydro-static test Press.	PD	PE	PF	PG	PK	Motor frame sizes a.c.	Motor frame sizes d.c.	PH	Group size shaft sleeve and water flinger
1C	1	1-1/4	4-3/4	3	3-1/4	75	3-1/2	2	2	17/32	3/8	56	56	5-7/8	B
2C	1	1-1/4	5 1/4	3-3/4	3-1/4	125	4-3/4	2-1/2	2-3/8	17/32	1/2	182TNC	186NC-187NC-216NC	6-1/4	S
3C	1-1/4	1-1/2	4-3/4	3-1/8	3-1/2	75	4-3/4	2-1/2	2-3/8	17/32	1/2	182TNC-184TNC-213TNC	216NC-218NC-254NC	6-1/4	S
							3-1/2	2	2	17/32	3/8	56	56	5-7/8	B
4C	1-1/2	2	5-1/2	3-3/4	3-3/4	100	4-3/4	2-1/2	2-3/8	17/32	1/2	182TNC-184TNC	186NC-187NC-216NC-218NC	6-1/4	S
5C	2	2	5-1/4	3-1/4	4	75	4-3/4	2-1/2	2-3/8	17/32	1/2	56 182TNC-184TNC 213TNC-215TNC	186NC-187NC-216NC-218NC 254NC-256NC	6-1/4	S
6C	3-1/2	3-1/2	7	5	4-1/2	50	4-3/4	2-1/2	2-3/8	17/32	1/2	182TNC-184TNC-213TNC	218NC-254NC-256NC	6-1/4	S
7C	1-1/4	2	6-3/4	5-1/4	4	250	7	4	4	21/32	3/4	254TNC-256TNC-286TNCI	284NC-286NC	8-3/16	M
8C	1-1/2	2-1/2	6-1/2	5	4	250						254TNC-256TNC-286TNCI	284NC-286NC		
9C	2-1/2	3-1/2	8	5-1/4	4-1/4	250						254TNC-256TNC-286TNCI	284NC-286NC-326NC		
10C	3-1/2	5	7-1/2	4-1/2	6	125						254TNC-256TNC-286TNCI	284NC-286NC-326NC		
11C	5	5	8-1/2	5-1/2	6	100						215TNC-254TNC-256TNC	284NC-286NC-326NC		
12C	6	6	9-1/2	8	6	50						254TNC-256TNC	286NC-324NC-326NC		
*13C	8	8	10-1/2	10	8-1/2	50	14	4	15-3/4	25/32	3/4	326TNCI	366NC-405NCI-444NCI	9-1/2	L
14C	1	1-1/4	6	4-1/4	3-1/2	200	4-3/4	2-1/2	2-3/8	17/32	1/2	213TNC	256NC	6-1/4	S
15C	2-1/2	3	6	3-3/4	4-1/2	75	4-3/4	2-1/2	2-3/8	17/32	1/2	213TNC-215TNC	254NC-256NC	6-1/4	S
*16C	6	8	9-1/2	8	8-1/2	100	14	4	14-1/2	25/32	3/4	286TNCI	365NCI-386NCI	9-1/2	L
17C	2-1/2	3	8	5-1/2	4-3/4	250	7	4	4	21/32	3/4		326NC	8-3/16	M

*Supporting feet part of volute casing or cover.

FIGURE 1. Outline dimensions for Navy standard close-coupled pumps.

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NOTES:

1. Radius shall be smooth and center locus concentric with shaft axis.
2. Sealing surface shall be smooth and perpendicular to shaft axis.
3. Diameter to be concentric with shaft at impeller location within 0.004 inch total indicator reading.
4. Face of flange to be square with shaft within 0.004 inch total indicator reading.
5. Runout of shaft 0.002 inch maximum total indicator reading.
6. Total axial end-play of shaft in assembled motor at no-load shall not exceed 0.010 inch. Bearing pre-load springs may be used to decrease end-play.

SH 2291A

STANDARD FRAME DIMENSIONS AND BEARING SIZES FOR FRACTIONAL HORSEPOWER CLOSE-COUPLED MOTORS

DIMENSIONS IN INCHES

FRAME NO.	A MAX	B MAX	D	E	F	H	BG	FRONT END BRG. (MIN.)	PUMP END BRG. (MIN.)	L MAX	MAX WT LB
56	6-3/4	11-1/2	3-1/2	2-7/16	1-1/2	17/32	4-1/16	303	305	9-1/4	55

TOLERANCES:

D Dimension Frame 56 ---------- +0", - 1/32"

2E and 2F ---------- +1/64", - 1/64"

BG Minus F ---------- +1/32", - 1/32"

Keys

Width 3/16" (.188)---------- +.000", - .002"

Depth

Shaft Diameter	Key Dimensions Width	Depth	Bottom of Keyway to Opposite Side of Shaft
.6875 ----	3/16 x	3/16 --------	.580 - .565

Keyways .188 ---------- +.001", - .001"

FIGURE 2. Standard dimensions of fractional hp close-coupled motors.

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TOLERANCES:

D Dimension

Frames 182 TCNI - 326 TCNI---------- +0", - 1/32"

2E and 2F ------- +1/64", - 1/64"

BG minus F ------- +1/32", - 1/32"

Keys

Width---------- +.000", - .002"

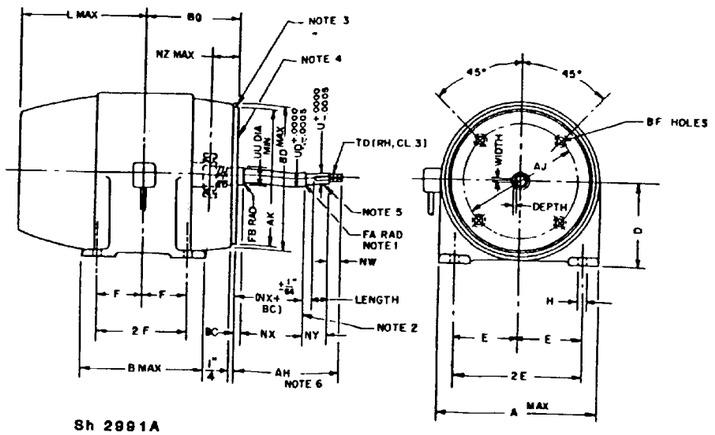
Depth

Shaft Diameter	Key Dimension Width	Depth	Bottom of Keyway To Opposite Side of Shaft
.6875 ----	3/16 x	3/16 -------	.580 - .565
1.0000 ----	1/4 x	1/4 -------	.859 - .844
1.3750 ----	3/8 x	3/8 ------	1.161 - 1.146

AK Dimension

10.5" and Smaller ---------- +.000", - .002"
12.5" and Larger ---------- +.000", - .003"

Keyways ---------- +.001", - .001"



Sh 2991A

NOTES:

1. Radius shall be smooth and center locus concentric with shaft axis.
2. Sealing surface shall be smooth and perpendicular to shaft axis.
3. Diameter to be concentric with shaft at "U" dimension within 0.004 inch total indicator reading.
4. Face of flange to be square with shaft within 0.004 inch total indicator reading.
5. The shaft runout shall be as follows:

Frame Size	Shaft Runout (TIR)
182 TCN thru 286 TCNI -----	0.002 inch max.
326 TCNI -----	0.003 inch max.

6. Total axial end-play of shaft in assembled motor at no-load shall not exceed 0.010 inch for .310 or smaller bearings and 0.015 inch for bearings larger than 310. Bearing pre-load springs may be used to decrease end-play.
7. All dimensions shown on Fig. 3 comply with MIL-M-17060, except for frames 286 TCNI and 326 TCNI which require larger bearings, shafts, and mounting bracket.
8. The "O" ring mating surface on the shaft shall have a 32 finish (see figure 8, detail A).

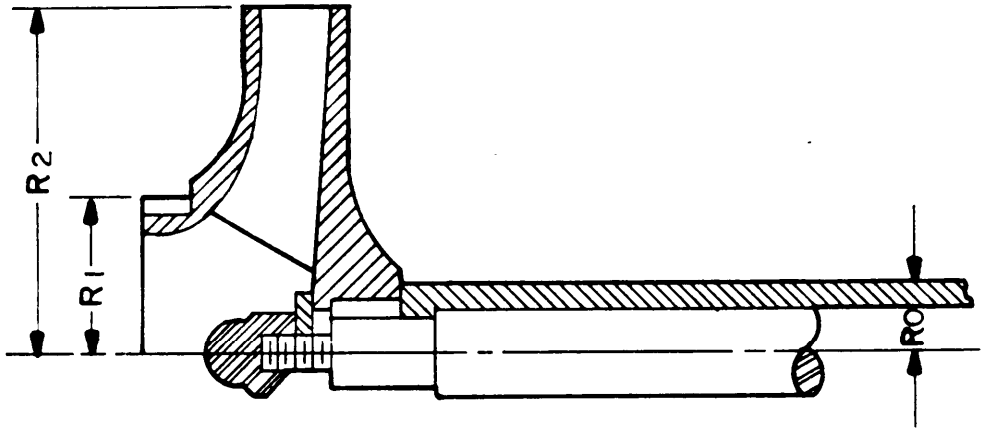
STANDARD FRAME FLANGE AND SHAFT EXTENSION DIMENSIONS AND BEARING SIZES FOR INTEGRAL HORSEPOWER CLOSE-COUPLED AC MOTORS

DIMENSIONS IN INCHES

FRAME NO.	A MAX	B MAX	D	E	F	H	BG	FRONT END BRG (MIN)	PUMP END BRG (MIN)	BD MAX	AJ	AX	BC	BF TAP	BF DEPTH	AH	NW	NY	NX	TD	U	UD	FA	FB	KEYWAY WIDTH	KEYWAY DEPTH	KEYWAY LGTH	KEY SQ. END	NZ MAX	UU MIN	L MAX	MAX WGT LBS	FRAME NO.
182 TCN	9	6-1/2	4-1/2	3-3/4	2-1/4	17/32	5-7/8	305	307	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-5/8	1.1875	9	105	182 TCN
184 TCN	9	7-1/2	4-1/2	3-3/4	2-3/4	17/32	6-3/8	305	307	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-5/8	1.1875	9-1/2	130	184 TCN
213 TCN	10-1/2	7-1/2	5-1/4	4-1/4	2-3/4	21/32	7-1/4	305	308	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-7/8	1.375	10-1/4	170	213 TCN
215 TCN	10-1/2	9	5-1/4	4-1/4	3-1/2	21/32	8	305	308	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-7/8	1.375	11	205	215 TCN
254 TCN	12-1/2	10-3/4	6-1/4	5	4-1/8	25/32	9-1/8	306	310	10	7-1/4	8-1/2	11-1/16	1/2-13	3/4	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	1/4	1/4	1/8	2	1-15/16	2-1/4	1.750	13	305	254 TCN
256 TCN	12-1/2	12-1/2	6-1/4	5	5	25/32	10	306	310	10	7-1/4	8-1/2	11-1/16	1/2-13	3/4	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	1/4	1/4	1/8	2	1-15/16	2-1/4	1.750	14	370	256 TCN
286 TCN	14	14	7	5-1/2	5-1/2	25/32	10-1/2	308	312	13	9	10-1/2	11-1/16	5/8-11	1	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	3/8	1/4	1/8	2	1-15/16	2-3/8	2.000	15-1/2	525	286 TCN
286 TCNI	18	15-1/2	9	7	5-5/8	1-1/16	11-3/4	309	313	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-11/16	2.000	17	970	286 TCNI
326 TCNI	18	15-1/2	9	7	5-5/8	1-1/16	11-3/4	309	313	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-11/16	2.000	17	970	326 TCNI

FIGURE 3. Standard dimensions and bearing sizes for integral hp close-coupled ac motors.

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SH 1319914

FIGURE 4. Outside of shaft sleeve.

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TOLERANCES:

D Dimension

Frames 186NC - 326NC ---------- +0", - 1/32"
Frames 364NC - 444NCH ---------- -0", - 1/16"

2E and 2F ---------- +1/64", - 1/64"

BG minus F ---------- +1/32", - 1/32"

Keys

Width ---------- +.000", - .002"

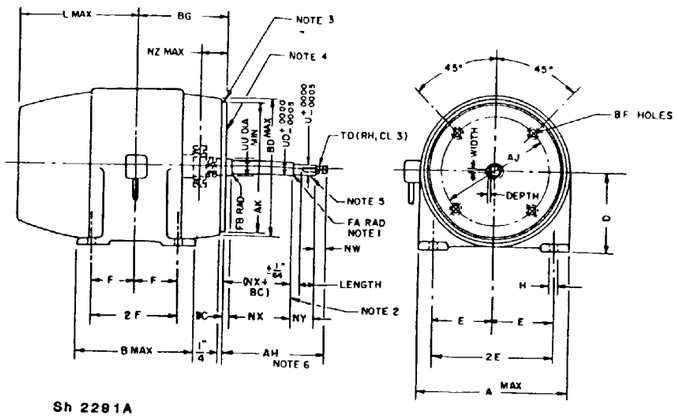
Depth

Shaft Diameter	Key Dimension Width	Depth	Bottom of Keyway To Opposite Side of Shaft
.6875 ----	3/16 x	3/16 -------	.580 - .565
1.0000 ----	1/4 x	1/4 -------	.859 - .844
1.3750 ----	3/8 x	3/8 -------	1.161 - 1.146

AK Dimension

10.5" and Smaller ---------- +.000", - .002"
12.5" and Larger ---------- +.000", - .003"

Keyways ---------- +.001", - .001"



Sh 2291A

NOTES:

1. Radius shall be smooth and center locus concentric with shaft axis.
2. Sealing surface shall be smooth and perpendicular to shaft axis.
3. Diameter to be concentric with shaft at "D" dimension within 0.004 inch total indicator reading.
4. Face of flange to be square with shaft within 0.004 inch total indicator reading.
5. The shaft runout shall be as follows:

Frame Size	Shaft Runout (TIR)
186 thru 286NC ------	0.002 inch max.
324NC thru 444NCH ---	0.003 inch max.

6. Total axial end-play of shaft in assembled motor at no-load shall not exceed 0.010 inch for 310 or smaller bearings and 0.015 inch for bearings larger than 310. Bearing pre-load springs may be used to decrease end-play.
7. The "O" ring mating surface on the shaft shall have a 32 finish (see figure 4, detail A).

STANDARD FRAME FLANGE AND SHAFT EXTENSION DIMENSIONS AND BEARING SIZES FOR INTEGRAL HORSEPOWER CLOSE-COUPLED DC MOTORS

DIMENSIONS IN INCHES

FRAME NO.	A MAX	B MAX	D	E	F	U	BG	FRONT END BRG (MIN)	REAR END BRG (MIN)	BD MAX	AJ	AK	BC	BF TAP	BF DEPTH	AH	NW	NY	NX	TD	U	UU	FA	FB	KEYWAY WIDTH	KEYWAY DEPTH	KEYWAY LGTH	KEY SQ. END	NZ MAX	UU MIN	L MAX	MAX WGT LBS	FRAME NO.
186NC	9	9	4-1/2	3-3/4	3-1/2	21/32	6-3/8	304	307	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-5/8	1.250	12-3/16	140	186NC
187NC	9	10	4-1/2	3-3/4	4	21/32	6-7/8	304	307	9	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-5/8	1.250	12-5/16	150	187NC
210NC	10-1/2	12	5-1/4	4-1/4	4	21/32	7-3/4	305	307	9	7-1/4	8-1/2	3/8	1/2-13	3/4	9-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-7/8	1.250	12-5/16	215	216NC
256NC	12-1/2	10-3/4	6-1/4	5	4-1/8	25/32	8-5/8	306	308	10	7-1/4	8-1/2	3/8	1/2-13	3/4	9-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	1-7/8	1.250	13-7/8	250	218NC
258NC	12-1/2	12-1/2	6-1/4	5	5	25/32	9-1/2	306	308	10	7-1/4	8-1/2	3/8	1/2-13	3/4	6-5/8	3/4	1-1/2	4	1/2-13	11/16	7/8	1/32	1/4	3/16	3/32	1-5/16	1-1/4	2-1/8	1.375	15-5/8	315	254NC
284NC	14	12-1/2	7	5-1/2	4-3/4	25/32	9-3/4	307	310	13	9	10-1/2	11/16	5/8-11	1	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	3/8	1/4	1/8	2	1-15/16	2-1/4	1.750	17-1/2	415	284NC
286NC	14	14	7	5-1/2	5-1/2	25/32	10-1/2	307	310	13	9	10-1/2	11/16	5/8-11	1	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	3/8	1/4	1/8	2	1-15/16	2-1/4	1.750	17-1/2	415	286NC
324NC	16	14	8	6-1/4	5-1/4	15/16	10-3/4	309	312	13	9	10-1/2	11/16	5/8-11	1	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	3/8	1/4	1/8	2	1-15/16	3	2.000	19-9/16	630	324NC
326NC	16	15-1/2	8	6-1/4	6	15/16	11-1/2	309	312	13	9	10-1/2	11/16	5/8-11	1	9-9/16	1	2-1/4	5-5/8	5/8-11	1	1-1/4	1/32	3/8	1/4	1/8	2	1-15/16	3	2.000	19-9/16	630	326NC
364NCH	18	16-1/4	9	7	6-1/8	1-1/16	12-1/4	310	313	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-11/16	2.000	20 5/16	685	326NC
366NCH	18	18	9	7	7	1-1/16	13	310	313	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-11/16	2.000	21-1/4	875	365NCH
405NCH	20	17-3/4	10	8	6-7/8	1-1/16	13-3/4	310	313	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-11/16	2.000	22-1/8	975	368NCH
444NCH	22	18-1/2	11	9	7-1/4	1-1/16	15	311	315	16-1/2	14	16	0	5/8-11	1	10-3/4	1-1/4	3	6-1/2	7/8-9	1-3/8	1-3/4	1/16	1/4	3/8	3/16	2-3/4	2-5/8	3-15/16	2.500	25-1/4	1625	444NCH

FIGURE 5. Showing dimensions and bearing sizes for integral hp close-coupled dc motors.

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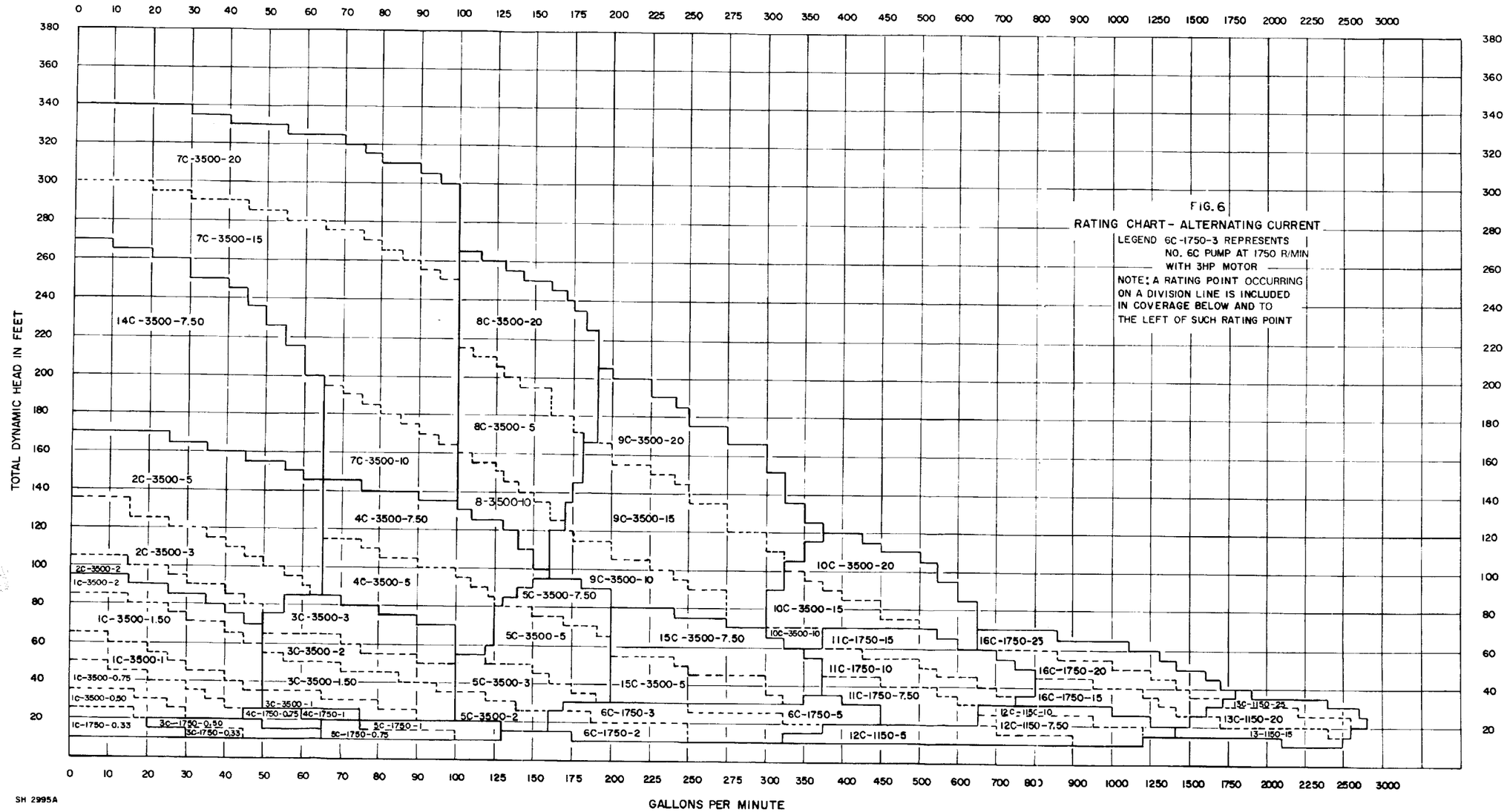


FIGURE 6. Rating chart for ac motors.

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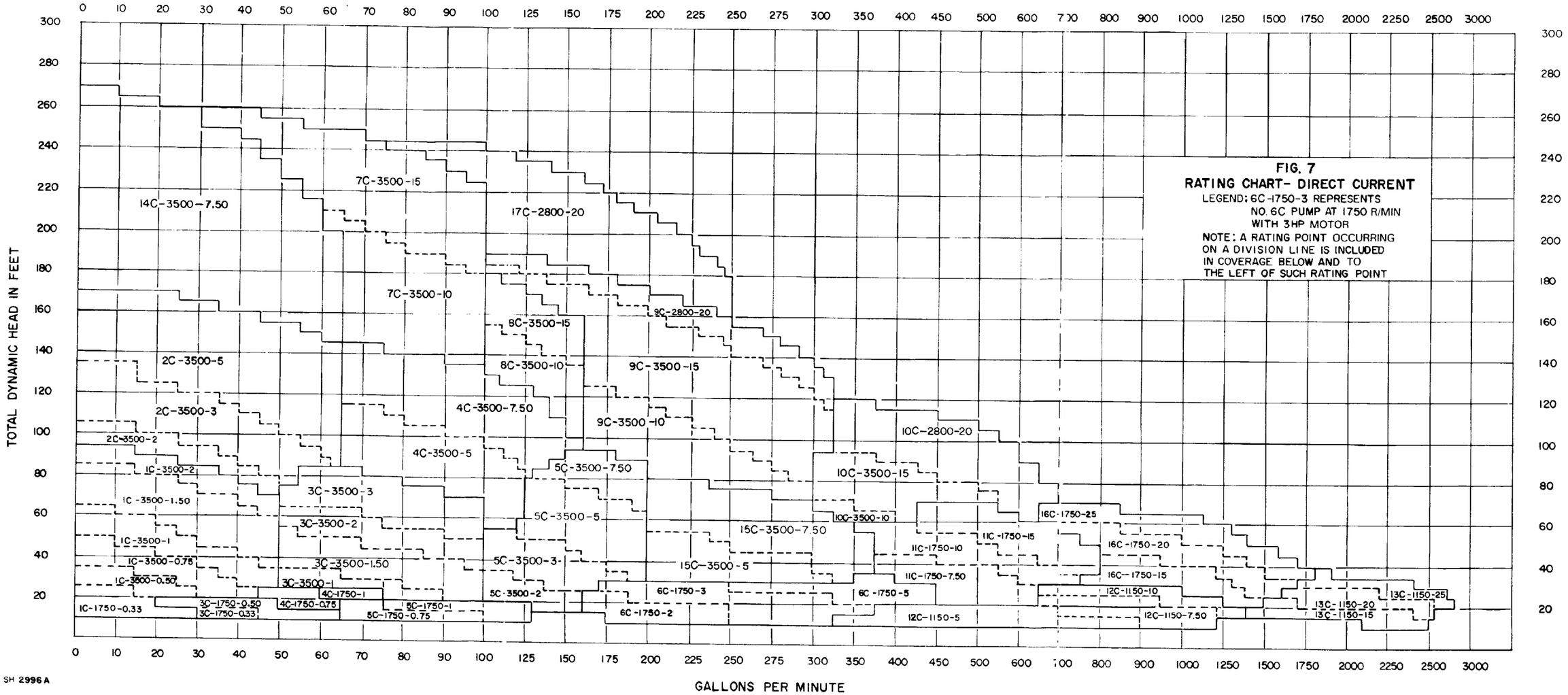
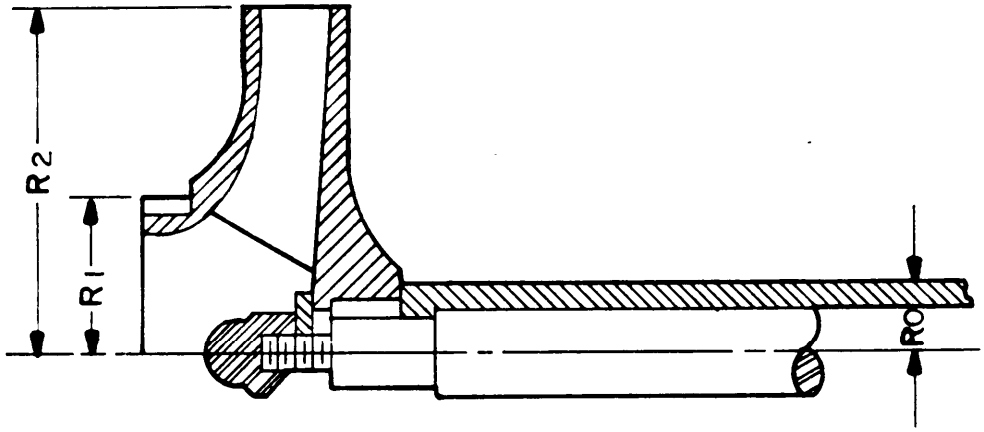


FIGURE 7. Rating chart for dc motors.

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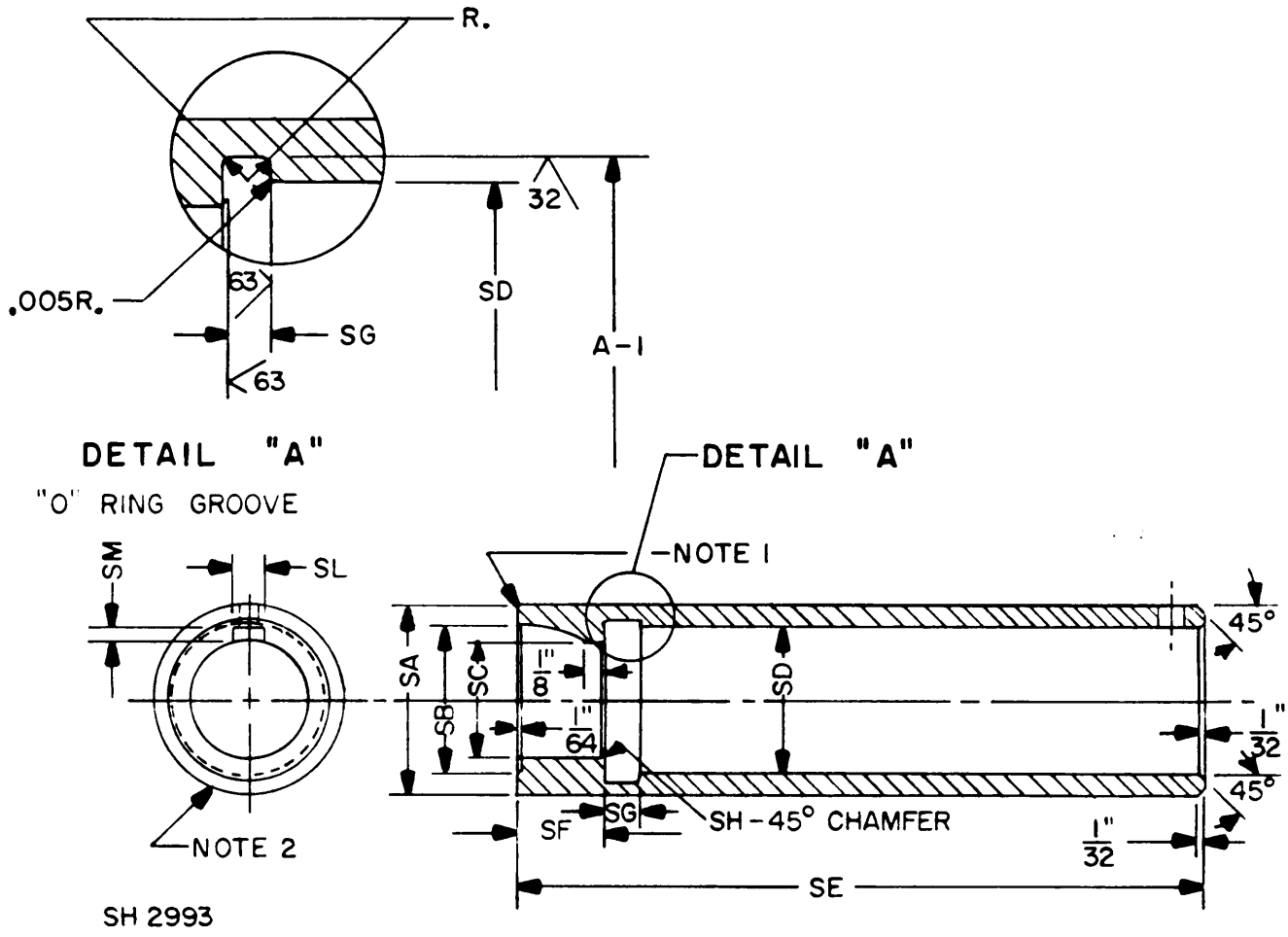


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FIGURE 4. Outside of shaft sleeve.

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GROUP	SA +.000 −.001	SB	SC +.002 −.000	SD +.001 −.000	SE	SF +.005 −.005	SG	SH	SL	SM	SR	ST	R	+.002 −.000 A-1	"O"-RING SIZE
S	1.125	15/16	.690	.8760	AS REQUIRED	.500	.093 .098	1/32	3/16	3/32	3/16	3/32	.005 .015	.975	2-020
M	1.625	1 5/16	1.003	1.2510		.625	.140 .145	1/32	1/4	1/8	1/4	13/64	.005 .015	1.412	2-124
L	2.250	1 7/8	1.378	1.7510		.750	.187 .192	1/16	3/8	3/16	5/16	1/4	.010 .025	1.972	2-224

NOTE 1: SEALING SURFACES SHALL BE SMOOTH TRUE AND PERPENDICULAR WITH BORE
NOTE 2: OUTSIDE DIAMETER SHALL BE GROUND SMOOTH TRUE AND CONCENTRIC WITH BORE

Figure 8. Shaft sleeve detail.

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